PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

Calma Transportation Services: Implementation of a Cash Register and Reservation System

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Maulion, John Welter

Moises, John Michael

Provido, Michael Harold

Cajayon, Renzo

Ramo, Carel Dame

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Transportation has always played a fundamental role in human society, serving as one of the essential systems that enable the movement of people, goods, and services from one location to another in an efficient, organized, and reliable manner. It is a critical component of economic development and social mobility, as it connects individuals to workplaces, educational institutions, healthcare services, and other essential destinations. In today's rapidly evolving and highly connected world, transportation has become even more significant due to the increasing demand for mobility, accessibility, and convenience. Modern lifestyles require faster and more flexible transportation options, especially in urbanized and tourism-driven areas where mobility is no longer considered a luxury but a daily necessity.

In particular, the demand for convenient and reliable transportation services continues to grow among various groups of individuals, including tourists, business professionals, students, and local residents. The increasing movement of people across cities and regions has created a need for transportation solutions that are both flexible and cost-effective. Many individuals prefer temporary access to vehicles rather than owning one due to financial constraints, maintenance costs, insurance expenses, and other long-term responsibilities associated with vehicle ownership. As a result, car rental services have emerged as a practical and efficient solution that allows users to access transportation based on their specific needs, schedules, and duration of use.

Car rental services provide a wide range of benefits such as flexibility, affordability, and convenience, making them an essential part of modern transportation systems. These services are particularly valuable in areas with high tourism activity, where visitors require short-term transportation to explore destinations, attend business meetings, participate in events, or travel between locations efficiently. Likewise, local residents may also rely on rental services for special occasions such as family trips, emergencies, or temporary vehicle replacement. Because of these growing demands, car rental businesses must ensure that their operations are well-organized, efficient, and capable of handling increasing customer requirements.

Car rental businesses operate by allowing customers to reserve and use vehicles for a specific period of time based on their transportation needs. This process typically involves vehicle selection, reservation scheduling, payment processing, and record management. While these steps may appear simple, they require accurate coordination to ensure that vehicles are properly allocated and available when needed. However, as customer demand increases, managing these operations becomes more complex and challenging, especially when handled manually or through traditional methods.

Efficient management of reservations, vehicle availability, customer records, and payment transactions is critical to the success and sustainability of any car rental business. Without proper systems in place, these processes can easily become disorganized, time-consuming, and prone to human error. Businesses must ensure that all transactions are properly recorded, updated, and monitored in real time to maintain smooth operations and high levels of customer satisfaction. Failure to implement an efficient system may result in operational inefficiencies, delayed services, booking conflicts, customer dissatisfaction, and potential financial losses.

Calma Transportation Services is a car rental business that provides vehicle booking and reservation services to both local customers and tourists. The company plays an important role in offering accessible and practical transportation solutions within its service area. It caters to individuals who require reliable transportation services for short-term use, ensuring that customers can travel conveniently without the need for vehicle ownership. However, despite the value of its services, the current operational process of Calma Transportation Services is primarily based on manual methods of recording and managing transactions.

At present, information such as customer bookings, vehicle availability, and payment records are typically documented using handwritten notes, logbooks, and basic record keeping systems. While these traditional methods may have been sufficient during the early stages of the business, they are no longer efficient in handling increasing transaction volumes. Manual processes require more time and effort, and they lack the ability to provide real-time updates, which are essential for modern business operations.

One of the major challenges of manual processing is the difficulty in maintaining accurate, consistent, and well-organized records. Handwritten entries are prone to human errors such as incorrect encoding, misinterpretation of information, and duplication of records. These issues become more frequent when multiple transactions occur at the same time, increasing the risk of booking conflicts where a single vehicle may be assigned to multiple customers. Such problems can cause delays, inconvenience, and dissatisfaction among clients, ultimately affecting the reputation of the business.

In addition, physical records are highly vulnerable to damage, loss, and misplacement. Unlike digital systems that store data securely in databases, manual records depend entirely on physical storage such as paper documents and logbooks. These materials can be easily affected by environmental factors such as moisture, fire, or physical wear and tear. Once records are lost or damaged, retrieving accurate information becomes difficult or even impossible, which disrupts business continuity and operational efficiency.

Another critical issue associated with manual systems is the complexity and inaccuracy of payment processing. Staff members are required to manually compute rental fees, additional charges, and discounts, which increases the likelihood of miscalculations. Even small errors in billing can lead to financial discrepancies, customer disputes, and loss of trust in the business. Moreover, generating reports for daily, weekly, or monthly transactions becomes a time-consuming process since data is not stored in a centralized and easily accessible system.

As technology continues to advance, many businesses across different industries are shifting from traditional manual processes to computerized and web-based systems. Digital transformation has become a key factor in improving operational efficiency, reducing human error, and enhancing data management capabilities. In the context of transportation and car rental services, computerized systems offer significant advantages such as real-time updates on vehicle availability, automated reservation processing, faster transaction handling, and improved financial accuracy through integrated cash register systems.

A web-based system, in particular, provides additional benefits in terms of accessibility, flexibility, and scalability. Through a web browser, authorized users such as staff and administrators can access the system anytime and anywhere, allowing for more convenient management of daily operations. This eliminates the limitations of traditional systems that require physical presence or manual checking of records. Furthermore, data is securely stored in a centralized database, ensuring that all records are properly organized, consistently updated, and easily retrievable when needed.

The integration of a reservation system with a cash register module further enhances the functionality of the business. This combination allows seamless coordination between booking management and payment processing, ensuring that all transactions are accurately recorded and properly monitored. It reduces redundancy, minimizes human error, and improves transparency in financial operations. In addition, it enables the generation of detailed reports that can be used for monitoring business performance, identifying trends, and supporting strategic decision-making.

The rapid advancement of information and communication technology has significantly transformed the way businesses operate across various industries. Organizations are increasingly adopting computerized systems to improve efficiency, accuracy, and accessibility in managing their daily operations. The transportation and vehicle rental industry is among the sectors that have greatly benefited from technological innovations. Through the implementation of web-based systems, businesses can automate routine processes, reduce paperwork, and provide faster services to customers. As technology continues to evolve, companies that embrace digital transformation gain a competitive advantage by improving service quality and operational performance.

In today's highly competitive business environment, customer satisfaction has become one of the most important factors influencing organizational success. Customers expect services that are fast, reliable, and accessible at any time. Traditional methods of managing reservations and transactions often fail to meet these expectations because they involve manual procedures that consume time and are prone to errors. As a result, many businesses are investing in digital solutions that allow customers to conveniently access services through online platforms. These systems improve communication between service providers and customers while enhancing the overall user experience.

The growing dependence on digital technologies has encouraged organizations to modernize their data management practices. Information serves as a valuable resource that supports decision-making and strategic planning. Businesses that maintain accurate and organized records are better positioned to monitor performance, identify trends, and respond to customer needs effectively. In contrast, organizations that continue to rely on manual record-keeping often experience challenges related to data inconsistency, information loss, and delayed reporting. Therefore, implementing a centralized information system is essential for ensuring the accuracy, security, and accessibility of business data.

The use of online reservation systems has become increasingly common in service-oriented industries because of their ability to streamline booking procedures and improve operational efficiency. Reservation systems provide customers with the convenience of checking availability, selecting services, and confirming bookings without the need for physical interaction. These systems also allow businesses to manage reservations more effectively by reducing scheduling conflicts and ensuring that resources are allocated appropriately. As customer expectations continue to rise, the adoption of reservation technologies becomes a necessity rather than an option for many organizations.

Financial management is another critical aspect of business operations that can benefit from automation. Accurate recording of transactions is essential for monitoring income, controlling expenses, and maintaining financial accountability. Cash register systems play a vital role in ensuring that all transactions are properly documented and easily retrievable when needed. By integrating financial management features into business systems, organizations can improve transparency, reduce errors, and generate accurate reports that support informed decision-making. The integration of cash register functionality with reservation management can further enhance operational efficiency by consolidating multiple processes into a single platform.

The emergence of web-based technologies has significantly changed the way businesses interact with their customers. Unlike traditional systems that require physical presence, web-based platforms allow users to access services remotely using computers or mobile devices connected to the internet. This level of accessibility provides greater convenience for customers while enabling businesses to extend their reach beyond geographical limitations. Furthermore, web-based systems facilitate real-time communication and information updates, ensuring that users receive accurate and timely information regarding available services and transactions.

As businesses continue to adapt to the digital era, the importance of system integration becomes increasingly evident. Integrated systems enable different business functions to work together seamlessly, reducing redundancy and improving efficiency. In the context of vehicle rental operations, integrating reservation management, customer records, vehicle monitoring, and financial transactions into a unified system can significantly enhance productivity. Such integration minimizes the need for repetitive data entry, improves information consistency, and supports more effective operational control.

The increasing demand for efficient transportation services has also contributed to the growth of vehicle rental businesses. Customers often seek flexible transportation options that provide convenience without the long-term responsibilities associated with vehicle ownership. As a result, car rental companies must continuously improve their services to meet customer expectations and remain competitive in the market. Implementing a modern information system allows rental businesses to respond more effectively to customer demands while ensuring efficient management of available resources.

The development of a Car Rental with Cash Register System aligns with the ongoing trend of digital transformation in the transportation industry. By utilizing modern technologies, the proposed system aims to improve reservation management, transaction processing, and record maintenance. The integration of these functionalities into a single platform can enhance operational efficiency, improve customer satisfaction, and support accurate financial management. Through automation and centralized data management, the system can address many of the challenges commonly associated with traditional rental operations.

The proposed system is expected to provide long-term benefits for both customers and administrators. Customers will experience a more convenient and efficient method of accessing rental services, while administrators will benefit from improved data organization and streamlined operations. Furthermore, the system can serve as a valuable tool for supporting business growth by providing accurate information and reliable reporting capabilities. As technology continues to shape modern business practices, the adoption of integrated information systems becomes increasingly important in achieving sustainable operational success.

The increasing dependence of businesses on digital technologies has transformed the manner in which services are delivered to customers. In recent years, organizations have recognized the importance of adopting computerized systems to remain competitive in rapidly changing markets. Technology has become an essential tool for improving operational efficiency, reducing human errors, and enhancing customer satisfaction. Businesses that utilize digital solutions are often able to provide faster, more accurate, and more reliable services compared to those that continue to rely on traditional manual procedures. As a result, information systems have become an integral component of modern business operations across various industries, including transportation and vehicle rental services.

The transportation sector plays a vital role in supporting economic growth and social mobility. People rely on transportation services for daily commuting, business activities, tourism, and other essential purposes. With the continuous increase in travel demands, vehicle rental businesses have emerged as a practical alternative for individuals who require temporary transportation without the financial burden of purchasing a vehicle. Car rental services provide flexibility and convenience, making them a popular option among tourists, business travelers, and local residents. Consequently, the demand for efficient and reliable rental management systems has increased significantly as businesses seek to accommodate growing customer expectations.

Many car rental businesses continue to face challenges related to reservation management, transaction recording, vehicle monitoring, and customer information management. Traditional methods often involve manual documentation using paper records, notebooks, and spreadsheets. While these methods may be effective for small-scale operations, they become increasingly difficult to manage as the volume of customers and transactions grows. Manual processes can result in inaccurate records, misplaced documents, delayed transaction processing, and difficulties in tracking vehicle availability. These operational issues can negatively affect business performance and reduce customer satisfaction.

One of the most common problems encountered by rental businesses is the occurrence of reservation conflicts. Reservation conflicts arise when multiple customers are assigned to the same vehicle due to inaccurate record keeping or delays in updating vehicle status. Such situations create inconvenience for customers and may damage the reputation of the business. Additionally, manual reservation processes often require employees to spend significant amounts of time verifying bookings, updating records, and communicating with customers. These repetitive tasks increase workload and reduce overall productivity. Implementing an automated reservation system can help eliminate these challenges by providing real-time updates and centralized management of booking information.

Customer satisfaction is considered a critical factor in determining the success of service-oriented businesses. Customers expect rental services to be efficient, reliable, and easily accessible. Delays in processing reservations, inaccurate information, and poor communication can negatively influence customer perceptions and discourage repeat business. Modern customers increasingly prefer online services that allow them to make reservations, review available options, and receive confirmations without visiting a physical office. Therefore, businesses must adopt technological solutions that provide convenience while ensuring accuracy and reliability throughout the rental process.

The emergence of web-based technologies has significantly influenced the development of modern business systems. Web-based applications enable users to access services from virtually any location with an internet connection. Unlike traditional desktop applications, web-based systems provide greater accessibility and flexibility for both customers and administrators. Customers can make reservations, check vehicle availability, and view transaction details using computers, tablets, or smartphones. Meanwhile, administrators can monitor business operations, manage records, and generate reports from a centralized platform. These capabilities contribute to improved efficiency and better overall service delivery.

Database management is another important aspect of modern information systems. A database serves as a centralized repository for storing, organizing, and retrieving information. In a car rental environment, databases can store customer information, vehicle records, reservation details, payment transactions, and operational reports. The use of a centralized database improves data accuracy by reducing duplication and ensuring consistency across different system functions. Furthermore, database systems facilitate faster retrieval of information, enabling employees to respond more effectively to customer inquiries and operational requirements.

Financial management is essential to the sustainability and growth of any business organization. Accurate recording and monitoring of financial transactions allow business owners to evaluate performance, identify trends, and make informed decisions. In many small rental businesses, financial records are still maintained manually, increasing the risk of calculation errors, missing receipts, and inaccurate reporting. Such issues can affect profitability and hinder effective decision-making. The integration of a cash register system into rental operations provides an efficient solution for managing financial transactions and ensuring accurate documentation of payments and revenues.

A cash register system automates the process of recording transactions, generating receipts, and maintaining financial records. Through automation, businesses can minimize human errors associated with manual computations and record keeping. Cash register systems also support transparency by providing detailed transaction histories that can be reviewed when necessary. In addition, these systems simplify report generation, enabling business owners to monitor daily, weekly, monthly, and annual financial performance. The availability of accurate financial information contributes to better planning and resource allocation within the organization.

The integration of a reservation management system with a cash register system provides numerous benefits for rental businesses. By combining these functions into a single platform, organizations can streamline operations and eliminate redundant processes. Reservation details can be automatically linked to payment records, reducing the need for duplicate data entry. This integration improves accuracy, enhances efficiency, and supports effective monitoring of business activities. Furthermore, it provides administrators with a comprehensive view of operational and financial information, enabling more informed decision-making.

As technological advancements continue to shape modern business environments, organizations are increasingly investing in digital transformation initiatives. Digital transformation involves the adoption of technologies that improve organizational processes, enhance customer experiences, and create new opportunities for growth. For car rental businesses, digital transformation may include online reservation systems, automated payment processing, vehicle tracking technologies, and data analytics tools. These innovations contribute to greater efficiency and competitiveness while addressing many of the limitations associated with traditional business practices.

The development of a Car Rental with Cash Register System represents an effort to address the operational challenges commonly experienced in vehicle rental businesses. By integrating reservation management and transaction processing into a unified web-based platform, the proposed system aims to improve efficiency, accuracy, and accessibility. The system seeks to provide customers with a convenient method of reserving vehicles while enabling administrators to manage bookings, monitor transactions, and generate reports more effectively. Through automation and centralized data management, the proposed system can support improved business performance and enhanced customer satisfaction.

In the long term, the adoption of automated rental management systems can contribute to the sustainability and growth of transportation service providers. Businesses that utilize technology effectively are better equipped to respond to changing customer demands and market conditions. Automated systems improve operational consistency, reduce administrative burdens, and provide valuable information for strategic planning. As the transportation industry continues to evolve, organizations that embrace technological innovation will be better positioned to achieve operational excellence and maintain a competitive advantage in the marketplace.

In response to these identified challenges, this study proposes the development of **"Calma Transportation Services: Cash Register and Reservation System."** The main objective of the proposed system is to replace existing manual processes with a fully computerized and automated solution that can streamline operations and improve overall service delivery. The system is designed to enhance the efficiency of reservation handling, improve the accuracy of payment processing, and ensure reliable data storage through a centralized database system.

By automating key processes such as booking management, payment computation, and record keeping, the system is expected to significantly reduce human errors, minimize operational delays, and improve overall productivity. It will also create a more organized workflow for staff members, allowing them to focus more on customer service and business improvement rather than time-consuming manual tasks.

Ultimately, the implementation of the proposed system will not only benefit the internal operations of Calma Transportation Services but will also greatly enhance the overall customer experience. Faster transaction processing, accurate booking records, improved system reliability, and better service coordination will contribute to higher customer satisfaction. In the long term, this digital transformation will support business growth, increase competitiveness, and ensure that Calma Transportation Services can effectively adapt to the evolving demands of modern transportation systems.

1.2 Statement of the Problem

Calma Transportation Services currently operates using manual methods in managing its vehicle reservations, customer information, and payment transactions. These processes rely heavily on handwritten logbooks, paper-based records, and manual encoding of transactions. While this traditional approach may have been sufficient during the early stages of operation when transaction volume was still limited, it has gradually become inefficient and increasingly difficult to manage as the number of customers and daily transactions continues to grow. The lack of automation in these processes limits the ability of the business to respond quickly and accurately to customer demands.

The continued reliance on manual systems presents several operational challenges that directly affect the efficiency, accuracy, and reliability of business operations. One of the primary concerns is the difficulty in monitoring vehicle availability in real time. Since records are updated manually, there is a higher possibility of outdated, incomplete, or inconsistent information being recorded. This can lead to scheduling conflicts, overlapping reservations, or double bookings, where more than one customer is assigned to the same vehicle at the same time. These issues not only disrupt daily operations but also negatively affect customer satisfaction, trust, and the overall reputation of the business. important business information. As transaction volumes increase, logbooks, receipts, and handwritten documents become harder to manage and more time-consuming to search through. Staff members may experience delays in retrieving booking histories, customer details, or payment records, especially when dealing with multiple transactions at once. This lack of efficient data organization can significantly slow down daily operations and reduce overall productivity and service efficiency.

Since all calculations are performed by hand, there is a higher risk of human error, especially when handling multiple transactions, discounts, additional fees, or varying rental durations. Even minor mistakes in computation can result in incorrect billing, financial discrepancies, and potential disputes between the business and its customers. Over time, these errors can affect the financial integrity of the business and reduce customer confidence in the accuracy of the services provided.

Furthermore, the absence of a centralized system makes it challenging to generate accurate, timely, and comprehensive reports that are essential for business monitoring and decision-making. Important data such as daily transactions, customer records, payment histories, and vehicle usage statistics are often scattered across different logbooks and physical files. This fragmented system makes it difficult for management to analyze business performance, identify trends, and evaluate operational efficiency. As a result, the business may struggle to make informed decisions that are necessary for improvement and growth.

Another concern is the lack of data security and backup in manual systems. Physical records are vulnerable to damage caused by environmental factors such as fire, water, wear and tear, or accidental misplacement. Once records are lost or damaged, recovering accurate information becomes extremely difficult or impossible. This poses a serious risk to the continuity and reliability of business operations, especially in cases where historical data is needed for verification or reporting purposes.

Given these challenges, there is a clear and growing need for a computerized reservation and cash register system that can enhance the efficiency, accuracy, and reliability of Calma Transportation Services. The development of such a system is expected to streamline business operations by automating key processes such as booking management, payment processing, and record keeping. It will also improve data organization by providing a centralized database that allows for real-time updates and easier retrieval of information.

Moreover, implementing a digital system will significantly reduce human error, improve transaction speed, and enhance overall service quality. Staff members will be able to perform their tasks more efficiently, while customers will benefit from faster and more accurate service delivery. Ultimately, the transition from a manual system to a computerized platform will not only modernize the operations of Calma Transportation Services but will also support its long-term growth, competitiveness, and ability to meet the increasing demands of the transportation industry.

Specifically, this study seeks to answer the following problems:

1. How can vehicle availability be monitored efficiently in real time to prevent double bookings and scheduling conflicts?
2. How can customer booking records be securely stored, organized, and easily retrieved when needed?
3. How can payment computations and cash register processes be automated to reduce errors in transactions?

4. How can the generation of reports and tracking of transaction history be improved to support better business monitoring and decision-making?

1.3 Objectives of the Study

1.3.1 General Objective

The main objective of this study is to design and develop a web-based Cash Register and Reservation System for Calma Transportation Services that will improve the efficiency, accuracy, reliability, and overall effectiveness of managing vehicle reservations, customer records, and payment transactions through a centralized and automated digital platform. The proposed system aims to replace the existing manual processes, which rely heavily on handwritten records and paper-based documentation, with a computerized solution that streamlines business operations and minimizes the occurrence of human errors.

Specifically, the system seeks to automate key operational activities such as reservation management, vehicle availability monitoring, customer information management, payment processing, and report generation. By integrating these functions into a single platform, the system will provide a more organized and efficient approach to handling daily business transactions. It is also intended to improve the accessibility, security, and accuracy of business records by storing information in a centralized database that can be easily updated, retrieved, and monitored when needed.

Furthermore, the proposed system aims to support better decision-making by providing accurate and timely information regarding reservations, vehicle utilization, customer transactions, and financial records. Through automation and digital record management, the system is expected to reduce administrative workload, enhance productivity, improve service quality, and contribute to the continuous growth and modernization of Calma Transportation Services. Ultimately, the development of the Cash Register and Reservation System seeks to provide a reliable technological solution that will strengthen operational performance, increase customer satisfaction, and support the long-term success of the business.

1.3.2 Specific Objectives

This study aims to achieve the following specific objectives:

- To design and develop a user-friendly system that simplifies and improves the vehicle reservation process, allowing customers and staff to manage bookings more efficiently and accurately.
- To automate the recording, updating, and management of customer booking information in order to reduce manual encoding errors and improve data consistency.
- To implement a real-time monitoring feature that allows users and administrators to track vehicle availability and status, helping to prevent scheduling conflicts and double bookings.
- To develop an automated cash register function that accurately computes rental fees and other related charges, minimizing human errors in financial transactions.

- To create a centralized and secure database that stores all booking, customer, and transaction records in an organized manner for easy retrieval and management.
- To design and implement a reporting feature that generates summaries of reservations and financial transactions, supporting business monitoring and decision-making processes.
- To improve the overall efficiency, reliability, and performance of Calma Transportation Services by integrating all processes into a unified automated system.

1.4 Scope and Delimitation

This study focuses on the design and development of a **web-based Cash Register and Reservation System for Calma Transportation Services**. The primary purpose of the system is to improve the efficiency, accuracy, and reliability of managing vehicle reservations, customer records, and financial transactions by transitioning from a manual and paper-based process to a computerized and automated platform. The proposed system aims to modernize the current operational workflow of the business by providing a more structured, organized, and efficient approach to handling daily transactions and service requests.

The scope of the system includes essential functionalities that are necessary for the smooth operation of a car rental business. These functionalities include vehicle management, customer booking and reservation recording, automated payment processing, and centralized storage of all transaction-related data. Through the system, administrators and staff will be able to add, update, and manage vehicle information, ensuring that records are always accurate and up to date. The reservation module will allow for efficient scheduling of vehicle rentals, reducing the possibility of conflicts and ensuring that all bookings are properly recorded and monitored.

In addition, the system will include a real-time vehicle availability monitoring feature, which provides updated information on whether a vehicle is available, reserved, or currently in use. This feature is essential in preventing double bookings and ensuring that customers receive accurate information regarding available transportation options. The system will also support automated computation of rental fees and charges, ensuring that all financial transactions are processed accurately and consistently without the need for manual calculations.

Furthermore, the system will be capable of generating basic reports related to reservations, customer transactions, and financial summaries. These reports will assist the business in monitoring daily operations, evaluating performance, and supporting decision-making processes. By having access to organized and structured data, management can better understand business trends and identify areas that require improvement.

The system will be developed as a **web-based application**, allowing it to be accessed through standard web browsers such as Google Chrome, Microsoft Edge, or other internet-enabled platforms. It can be used on devices such as desktop computers, laptops, and other compatible devices connected to the internet or local network. This web-based approach ensures flexibility and convenience for users, as it eliminates the need for installation on multiple devices and allows centralized access to system functions.

The primary users of the system will be the **business owner and authorized staff members** of Calma Transportation Services. These users are responsible for managing reservations, handling customer information, processing payments, and maintaining vehicle records. The system is designed to assist them in performing their tasks more efficiently by reducing manual workload, minimizing repetitive processes, and decreasing the likelihood of human errors. As a result, staff members can focus more on customer service and business improvement rather than administrative tasks.

However, despite the advantages of the proposed system, the study also has certain limitations and delimitations. The system will not include advanced functionalities such as GPS tracking of vehicles, integration with external online payment gateways such as GCash, PayPal, or credit card processing systems, or the development of a dedicated mobile application. These features are excluded due to scope limitations and the focus on core reservation and cash register functionalities.

Additionally, the system will not support full-scale enterprise deployment and will be developed primarily as an academic prototype intended for research, demonstration, and evaluation purposes. As such, it is designed to simulate real-world operations but may still require further enhancement, optimization, and testing before being implemented in a full commercial environment. Future improvements may include system scalability, enhanced security features, and integration with advanced technologies depending on business needs and further development opportunities.

1.5 Significance of the Study

This study is expected to provide meaningful contributions and benefits to various stakeholders, particularly in improving the efficiency and effectiveness of transportation and reservation management processes within Calma Transportation Services. By developing a computerized Cash Register and Reservation System, the study aims to address existing operational challenges and enhance overall service delivery.

Calma Transportation Services

The primary beneficiary of this study is Calma Transportation Services. The proposed system will significantly improve operational efficiency by automating key business processes such as vehicle reservations, customer record management, and payment transactions. This reduces reliance on manual procedures, minimizes human errors, and ensures more accurate and organized data handling. As a result, the business will be able to provide faster services, improve workflow management, and enhance overall productivity and profitability.

Employees / Staff

The employees or staff members of Calma Transportation Services will benefit from the system through simplified and more efficient work processes. The system provides a centralized and user-friendly platform that allows staff to easily manage reservations, monitor vehicle availability, and process payments. This reduces workload, eliminates repetitive manual tasks, and helps employees focus more on customer service and operational improvement rather than administrative difficulties.

Customers

Customers will greatly benefit from the improved speed, accuracy, and reliability of services. With the implementation of the system, booking transactions become faster and more convenient, reducing waiting time and eliminating common issues such as double bookings or incorrect reservation details. This leads to a more satisfying customer experience and improves trust and confidence in the services provided by Calma Transportation Services.

Researchers / Developers

For the researchers and developers, this study serves as an opportunity to enhance their technical knowledge and practical skills in system development, programming, database design, and system integration. It also allows them to apply theoretical concepts learned in academic settings into a real-world application. Through this process, the researchers gain valuable experience in developing a functional and structured information system.

Future Researchers

This study will serve as a useful reference for future researchers who intend to explore similar topics related to reservation systems, transportation management systems, and business process automation. It may provide insights into system design, development methodologies, and implementation strategies that can be adapted or improved in future studies. Additionally, it can serve as a baseline for further enhancements or more advanced system development projects.

1.6 Definition of Terms

To provide clarity and a better understanding of the key concepts used in this study, the following terms are defined as used in the context of the system:

Booking – The act of reserving a vehicle for use at a specific date and time through the system. It represents a confirmed request made by a customer to secure transportation services. In this study, booking ensures that each reservation is properly recorded, scheduled, and monitored to avoid conflicts such as double bookings or overlapping schedules.

Cash Register – A digital feature of the system that automatically records, computes, and manages payment transactions related to vehicle rentals. It ensures accurate and efficient handling of financial records by reducing manual computation errors and providing real-time transaction summaries for the business.

Customer – An individual who avails or reserves a vehicle from Calma Transportation Services through the reservation system. Customers may include tourists, local residents, or business clients who require transportation services for a specific period of time.

Database – A structured and centralized electronic system used for storing, organizing, managing, and retrieving all relevant data such as customer information, reservations, vehicle details, and financial transactions. It ensures data integrity, consistency, and security within the system.

Reservation – An advance arrangement made by a customer through the system to secure a vehicle for a specific date and time of use. Reservations help ensure that vehicles are properly scheduled and allocated based on customer demand and availability.

Transaction – Any activity recorded within the system that involves booking, reservation, payment, or other service-related processes. Transactions serve as official records of business operations and are essential for monitoring financial performance.

Vehicle Availability – The current status of a vehicle indicating whether it is available for use, reserved, or currently rented. This feature is important in managing fleet operations and preventing scheduling conflicts within the system.

Web-Based System – An application that operates through a web browser and can be accessed via the internet or local network. It allows users such as staff and administrators to manage reservations, payments, and records efficiently without requiring installation on a specific device.

Automation – The use of technology to perform processes automatically with minimal human intervention. In this study, automation is applied in booking management, payment computation, and record handling to improve efficiency and reduce human errors.

Calma Transportation Services – The business organization where the proposed system will be implemented. It is a transportation service provider that offers vehicle rental and reservation services to customers and currently operates using manual processes.

User Interface (UI) – The visual layout and interactive components of the system that allow users to navigate and perform tasks such as booking, payment processing, and record management. A well-designed UI improves usability and enhances user experience.

System – Refers to the proposed **Calma Transportation Services: Cash Register and Reservation System**, which is a web-based application designed to automate and improve the management of bookings, payments, and transportation operations.

Real-Time Processing – The ability of the system to update and display information instantly as changes occur, such as updating vehicle availability after a booking or payment transaction.

Record Management – The process of organizing, storing, and retrieving data within the system to ensure that all customer, booking, and financial records are properly maintained and easily accessible.

Efficiency – Refers to the ability of the system to perform tasks quickly and accurately while minimizing time, effort, and resource usage in managing reservations and transactions.

Accuracy – The degree to which the system produces correct and error-free results, particularly in booking schedules, payment computations, and record-keeping.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

The rapid advancement of information technology has significantly transformed the way services are delivered and managed across various industries worldwide, particularly in transportation and vehicle rental services. As businesses continue to adopt digital solutions, traditional manual processes are gradually being replaced by web-based systems that offer faster, more accurate, and more efficient methods of handling transactions and operations. This shift toward digitalization is driven by the growing demand for convenience, real-time access to information, and improved service quality.

In the context of vehicle rental services, foreign studies have shown that online car rental systems play a vital role in enhancing operational efficiency by automating key processes such as reservation handling, vehicle availability tracking, payment processing, and customer management. These systems reduce the reliance on manual documentation and face-to-face transactions, which are often prone to delays, errors, and inconsistencies. Instead, web-based platforms allow users to access services anytime and anywhere, improving accessibility and overall user experience.

Moreover, foreign literature highlights that the integration of modern technologies such as real-time databases, mobile applications, automated notifications, and intelligent system features contributes significantly to improving customer satisfaction and business performance. These systems not only streamline internal operations but also promote transparency, reliability, and better communication between service providers and customers.

Overall, foreign studies consistently emphasize that the adoption of web-based car rental systems is essential in modernizing transportation services. They demonstrate that digital transformation leads to improved efficiency, reduced operational costs, and enhanced service delivery, making it a critical component in the development of modern car rental management systems.

Osman et al. (2017) developed a web-based car rental system integrated with SMS notification technology to enhance communication efficiency between rental companies and customers. The researchers identified that traditional car rental operations often experience communication delays, limited real-time updates, and inefficiencies in informing customers about reservation status and vehicle availability. These issues can lead to customer dissatisfaction, confusion, and reduced trust in rental services, especially when confirmations and updates are not delivered promptly.

To address these problems, the researchers designed a web-based platform that allows customers to reserve vehicles online while receiving automated SMS notifications regarding booking confirmations, reservation status, and rental updates. The system integrates a centralized database with a communication module that ensures real-time dissemination of information between the system and users. This structure enables continuous updates without requiring manual intervention from staff, improving both speed and accuracy of communication.

The findings of the study showed that the integration of SMS notification technology significantly improved transparency and customer trust, as users were consistently informed about the status of their transactions. In addition, the system reduced the need for customers to contact rental offices for inquiries, resulting in improved convenience and reduced workload for administrative staff. The researchers also noted that automation of notifications minimized human error and improved the overall efficiency of the rental process.

The study concluded that integrating communication technologies such as SMS into web-based rental systems plays a vital role in improving operational efficiency, service reliability, and customer satisfaction. This study is relevant to the proposed Car Rental with Cash Register System because it highlights the importance of real-time communication and automated transaction updates in ensuring smooth operations and enhanced customer experience.

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Source: https://jcrinn.com/index.php/jcrinn/article/view/51

Azmi and Hassan (2024) introduced a web-based automobile rental system designed to enhance the efficiency and accessibility of vehicle rental services through digital automation. The researchers identified that many traditional car rental businesses still rely on manual processes such as walk-in reservations, paper-based records, and phone-based booking systems. These conventional methods often lead to operational inefficiencies, including booking conflicts, data entry errors, and delays in processing customer requests. To address these issues, the researchers developed a centralized web-based platform that allows users to browse available vehicles, create accounts, and manage rental reservations online.

The system was designed with both customer and administrator functionalities to ensure efficient operation. Customers can view vehicle availability, select preferred units, and submit reservation requests without the need to physically visit the rental office. On the administrative side, the system enables staff to monitor bookings, manage vehicle inventories, and update rental status in real time. This dual functionality improves coordination between users and service providers while reducing dependence on manual documentation.

The findings of the study revealed that the implementation of a web-based rental system significantly improves operational efficiency by reducing paperwork and minimizing human errors in data handling. It also enhances customer satisfaction by providing 24/7 access to rental services and real-time information updates. In addition, the automation of processes allows employees to focus more on customer service rather than repetitive administrative tasks. The researchers concluded that digital transformation is essential in modern car rental systems to improve service quality, accuracy, and overall business productivity. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how integrating online reservation features and automated record management can improve transaction efficiency and system reliability.

ISSN: 2773-5141

Source: https://periodical.uthm.edu.my/index.php/aitcs/article/view/12001?

Harissa (2025) developed a web-based car rental system using the Agile Scrum methodology to improve the efficiency, flexibility, and responsiveness of rental service operations. The researcher identified that traditional car rental systems often suffer from slow development cycles, inefficient booking processes, and limited adaptability to changing user requirements. These challenges can lead to outdated system features, delayed updates, and reduced customer satisfaction. To address these issues, the study adopted Agile Scrum, a software development approach that emphasizes iterative development, continuous feedback, and incremental system improvement.

The system developed in the study integrated essential car rental functionalities such as vehicle booking, reservation management, user authentication, and payment processing. By using Agile Scrum, the development process was divided into multiple sprints, allowing the system to be improved step-by-step based on user feedback and testing results. This approach ensured that issues were identified early and resolved efficiently, resulting in a more stable and user-centered application.

The findings of the study revealed that the integration of booking and payment functionalities into a single platform significantly improved transaction speed and reduced processing delays. In addition, the use of Agile methodology enhanced system adaptability, allowing developers to easily implement changes based on user needs and system performance evaluations. Customers also benefited from a smoother and more convenient booking experience due to the streamlined interface and automated processes.

The researcher concluded that applying Agile Scrum in the development of car rental systems leads to improved system quality, faster development cycles, and better user satisfaction. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how integrating booking, payment, and iterative system development can enhance operational efficiency and improve overall service delivery.

ISSN: P-ISSN 2527-9866, E-ISSN 2549-3469

Source: https://doi.org/10.35314/wfaayr38

Yang et al. (2021) focused on demand forecasting in car rental services using data-driven analytical models to improve operational planning and resource allocation. The researchers identified that car rental companies often face challenges in balancing vehicle supply and customer demand, especially during peak seasons. These imbalances may lead to either vehicle shortages, which reduce customer satisfaction, or excess fleet availability, which increases operational costs. To address this issue, the study proposed the use of predictive analytics techniques that analyze historical rental data, seasonal trends, and customer behavior patterns to forecast future demand more accurately.

The model developed in the study utilized statistical and computational approaches to generate predictions that support decision-making in fleet management. By analyzing previous rental records, the system can estimate which types of vehicles are likely to be in higher demand and at what specific time periods. This allows rental companies to adjust their fleet distribution, improve scheduling efficiency, and reduce unnecessary operational expenses. The researchers also emphasized that integrating data analytics into rental systems enables more informed decision-making, as managers can rely on evidence-based insights rather than manual estimation.

The findings of the study showed that accurate demand forecasting significantly improves operational efficiency by reducing idle vehicles and ensuring better availability of resources. In addition, the use of predictive models enhances customer satisfaction by minimizing the chances of unavailability during peak demand periods. The study concluded that data-driven forecasting is a valuable tool for improving competitiveness and sustainability in the car rental industry.

This study is relevant to the proposed Car Rental with Cash Register System because it highlights the importance of data management, reporting, and analytical capabilities in improving operational planning and supporting efficient business decision-making processes.

ISSN: 2076-3417

Source:

https://scholar.google.com/scholar?q=Unconstrained+Estimation+of+Multitype+Car+Rental+Demand

García-Moreno et al. (2022) explored the application of blockchain technology in vehicle rental systems by developing a decentralized platform that utilizes Ethereum smart contracts to automate rental transactions. The researchers identified that traditional car rental systems often rely on centralized databases and intermediaries, which may introduce issues related to trust, data manipulation, and transaction transparency. These limitations can result in disputes between customers and service providers, as well as inefficiencies in verifying rental agreements and payment processes. To address these challenges, the study proposed a blockchain-based system that enables secure, transparent, and automated execution of rental contracts without the need for intermediaries.

The system uses smart contracts to manage key rental operations such as vehicle booking, payment validation, and contract enforcement. Once predefined conditions are met, transactions are automatically executed and recorded on the blockchain, ensuring that all records are immutable and verifiable. This approach enhances data integrity because once information is stored in the blockchain, it cannot be easily altered or tampered with. As a result, both customers and rental providers can trust the accuracy and authenticity of transaction records.

The findings of the study revealed that blockchain integration significantly improves transparency, security, and trust in car rental systems. It also reduces administrative overhead by automating processes that would normally require manual verification and third-party involvement. Additionally, the system enhances accountability by providing a permanent and traceable record of all rental activities.

The researchers concluded that blockchain technology has strong potential to transform the car rental industry by improving operational security and transaction reliability. This study is relevant to the proposed Car Rental with Cash Register System because it highlights the importance of secure transaction processing, accurate record-keeping, and trustworthy financial management in modern rental systems.

ISSN: 1368-9894

Source: https://doi.org/10.1093/jigpal/jzac003

Ogbiti and Aaron (2024) developed a web-based car rental management system aimed at improving the efficiency and accuracy of vehicle rental operations through automation and centralized data management. The researchers identified that many traditional car rental businesses continue to depend on manual processes for handling reservations, customer records, and vehicle tracking. These manual methods often lead to delays in processing transactions, inconsistencies in data storage, and difficulties in monitoring vehicle availability in real time. Such challenges can negatively affect service quality and reduce customer satisfaction.

To address these issues, the researchers designed a web-based system that integrates key functionalities such as user registration, vehicle reservation, rental transaction management, and administrative monitoring tools. The system was developed to provide both customers and administrators with a centralized platform where rental activities can be efficiently managed.

Customers are able to view available vehicles, make reservations, and monitor booking status, while administrators can oversee all transactions, manage vehicle listings, and generate reports for decision-making purposes.

The findings of the study revealed that the implementation of the system significantly improved operational efficiency by reducing the time required to process reservations and update rental records. In addition, the use of a centralized database minimized data redundancy and reduced the likelihood of human errors commonly associated with manual documentation. The system also improved accessibility, allowing users to interact with the rental service more conveniently through an online platform.

The researchers concluded that web-based car rental systems provide a practical solution for modernizing rental operations by enhancing efficiency, accuracy, and customer satisfaction. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how automation and centralized transaction processing can improve operational performance and ensure reliable financial and record management.

ISSN: 2756-391X

https://doi.org/10.4314/swj.v19i3.27

Muttaqin, Wadly, and Veranda (2025) examined the development of a web-based car rental information system aimed at supporting the digital transformation of rental service operations. The researchers identified that many car rental businesses still rely on conventional methods such as manual booking forms, physical record keeping, and direct customer transactions. These traditional processes often result in inefficiencies, including slow reservation handling, data redundancy, and difficulties in tracking vehicle availability in real time. As customer demand for faster and more convenient services increases, these limitations become more significant and can negatively affect business performance and customer satisfaction.

To address these challenges, the researchers developed a web-based system that centralizes rental operations into a single digital platform. The system includes core functionalities such as user registration, vehicle listing management, reservation processing, and rental transaction tracking. It also provides administrative tools that allow staff to monitor bookings, update vehicle status, and generate operational reports. By consolidating these processes into one system, the platform improves coordination between customers and administrators while reducing dependence on manual documentation.

The study found that implementing a web-based information system significantly enhances operational efficiency by reducing processing time and minimizing human errors in data handling. It also improves data accuracy through centralized storage, ensuring that vehicle and transaction records are consistently updated and easily accessible. Furthermore, the system increases customer satisfaction by allowing users to access rental services anytime and anywhere without the need for physical interaction.

The researchers concluded that digital transformation plays a crucial role in improving the competitiveness and efficiency of car rental businesses. This study is relevant to the proposed Car Rental with Cash Register System because it highlights the importance of integrating automated reservation management, centralized databases, and real-time transaction monitoring to improve service delivery and operational reliability.

ISSN: 3046-4900

Source: https://ysmk.org/ejournal/index.php/jitcse/article/view/245

Rahman and Rusli (2023) developed a web-based car rental business management system designed to improve the overall efficiency, accuracy, and accessibility of rental operations. The researchers identified that many car rental businesses still rely on traditional processes such as manual reservation handling, paper-based record keeping, and in-person transactions. These conventional methods often lead to operational inefficiencies, including delayed booking confirmations, data inconsistencies, and difficulties in tracking vehicle availability. As a result, both customers and administrators experience challenges in managing rental transactions effectively.

To address these issues, the researchers designed a centralized web-based system that integrates key rental processes into a single platform. The system allows customers to view available vehicles, make reservations, and manage bookings online without needing to visit the rental office physically. On the administrative side, the system provides tools for managing customer records, monitoring reservations, updating vehicle availability, and generating business reports. This dual-function structure ensures smoother coordination between users and service providers while reducing reliance on manual documentation.

The study found that the implementation of the system significantly improved operational efficiency by reducing processing time and minimizing human errors associated with manual data entry. In addition, the centralized database enhanced data accuracy and ensured that information regarding vehicles and reservations remained consistent and up to date. The system also improved customer satisfaction by providing convenient, real-time access to rental services and reducing the need for physical transactions.

The researchers concluded that web-based systems are essential for modernizing car rental businesses and improving overall service delivery. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how integrating online reservation management, centralized databases, and automated transaction processing can enhance operational performance and support accurate financial and business record management.

ISSN:0128-0767

Source: https://doi.org/10.24191/jmcs.v9i2.513

Oktaviani and Sari (2025) developed a web-based car rental information system for CV. Rafael Trans with the aim of improving the efficiency, accuracy, and accessibility of rental operations. The researchers identified that traditional car rental processes in many small and medium-sized enterprises often rely on manual methods such as handwritten records, walk-in reservations, and direct communication through phone calls or messages. These approaches frequently result in operational issues such as double bookings, inaccurate data recording, and delays in processing customer requests, which negatively affect service quality and customer satisfaction.

To address these challenges, the researchers designed a web-based system that centralizes all rental processes into a single platform. The system allows customers to view available vehicles, register accounts, and make reservations online without needing to visit the rental office. On the administrative side, the system provides tools for managing vehicle data, monitoring reservations, updating rental status, and generating transaction reports. This structure improves coordination between customers and staff while ensuring that all rental information is stored in a centralized database.

The findings of the study revealed that the implementation of the system significantly improved operational efficiency by reducing manual documentation and speeding up transaction processing. It also enhanced data accuracy by minimizing human errors commonly associated with traditional record-keeping methods. In addition, the system improved customer satisfaction by providing 24/7 access to rental services and real-time updates on vehicle availability.

The researchers concluded that web-based car rental information systems are effective tools for modernizing rental businesses and improving overall service delivery. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how integrating online reservation management, centralized databases, and automated reporting functions can enhance operational performance and ensure accurate transaction and financial record management.

P-ISSN: 1979-5246

Source: https://ejournal.usni.ac.id/index.php/jisni/article/view/3

Karsa et al. (2024) conducted a study on the development of a web-based car rental information system for CV. Mandarental Cirebon with the objective of improving service efficiency and addressing issues in manual rental management. The researchers identified that traditional car rental operations often rely on handwritten logs, direct customer inquiries, and manual reservation tracking. These practices frequently lead to problems such as inaccurate data recording, booking conflicts, and delays in updating vehicle availability. As a result, customers may experience inconvenience, while administrators struggle with inefficient workflow and increased workload.

To address these challenges, the researchers developed a web-based system that centralizes all rental processes into a single platform. The system enables customers to view available vehicles, submit reservation requests, and monitor booking status through an online interface. On the administrative side, it provides functionalities for managing vehicle data, confirming reservations, updating rental status, and generating reports for operational monitoring. The integration of these features into one system allows for better coordination between users and staff while ensuring that all records are stored in a centralized database.

The findings of the study showed that the implementation of the system significantly improved operational efficiency by reducing manual processing time and minimizing errors in data handling. It also enhanced data consistency and reliability by eliminating redundant record-keeping practices. Furthermore, the system improved customer experience by allowing faster access to rental services and providing real-time updates on vehicle availability.

The researchers concluded that web-based car rental information systems play a crucial role in modernizing rental businesses and improving service quality. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how automation, centralized databases, and online reservation features can enhance operational performance and ensure accurate transaction and record management.

ISSN: 2980-2841 Source: https://ajesh.ph/index.php/gp/article/view/246

Sari and Jihad (2024) developed a web-based car rental service portal information system aimed at improving the efficiency, accessibility, and accuracy of vehicle rental operations. The researchers identified that many car rental businesses still depend on traditional processes such as manual reservation forms, phone-based bookings, and in-person transactions. These conventional methods often lead to inefficiencies such as delayed processing of reservations, data entry errors, and difficulties in tracking vehicle availability in real time. Such limitations negatively affect both customer experience and operational performance.

To address these issues, the researchers designed a centralized web-based portal that integrates key rental processes into a single system. The platform allows customers to register accounts, browse available vehicles, submit reservation requests, and monitor rental status through an online interface. On the administrative side, the system provides tools for managing customer records, updating vehicle availability, processing reservations, and generating reports.

This integrated approach ensures that all rental-related data is stored in a centralized database, improving coordination between users and administrators.

The study found that the implementation of the system significantly improved operational efficiency by reducing manual workload and minimizing errors associated with traditional documentation. It also enhanced data accuracy by ensuring that all transactions and rental records are updated in real time. In addition, the system improved customer satisfaction by providing convenient access to rental services anytime and anywhere without the need for physical interaction.

The researchers concluded that web-based car rental portals are effective solutions for modernizing rental operations and improving overall service delivery. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how integrating reservation management, centralized databases, and automated reporting functions can improve operational efficiency and support accurate transaction processing and financial record management.

ISSN: 2964-9329

Source: https://ejournal.isha.or.id/index.php/Mandiri/article/view/467

Nugraha, Nurwanto, and Erwantoro (2022) developed a web-based vehicle rental application using PHP and MySQL to improve the efficiency of rental operations and data management. The researchers identified that manual rental processes often cause delays in recording transactions, difficulties in managing vehicle information, and errors in maintaining customer records. These challenges affect the accuracy of rental operations and reduce the quality of customer service.

To address these issues, the researchers designed a web-based application that automates customer registration, vehicle reservation, rental transactions, and report generation. The system uses PHP as the server-side programming language and MySQL as the database management system to store rental information in a centralized database. It also provides an administrative interface that enables staff to monitor vehicle availability, update rental records, and manage customer information more efficiently.

The findings showed that the implementation of the web-based application improved the efficiency of rental operations by reducing manual work, minimizing data entry errors, and speeding up transaction processing. The centralized database also improved the consistency and accessibility of rental information, allowing staff to manage operations more effectively while providing better service to customers.

The researchers concluded that implementing a web-based rental management system is an effective solution for improving operational efficiency, data accuracy, and customer service. This study is relevant to the proposed **Car Rental with Cash Register System** because it demonstrates how integrating automated reservation processing, centralized database management, and digital transaction recording can improve rental operations and strengthen financial record management.

ISSN: 2714-5417

Source: https://journal.stekom.ac.id/index.php/elkom/article/view/857

Ahmed and Yunus (2023) developed the AL-WEFAQ Car Rental Reservation System to address the limitations of traditional car rental businesses that rely heavily on manual processes and face challenges in reaching customers effectively. The researchers observed that many rental companies struggle with inefficient reservation handling, limited online presence, and difficulty in managing customer bookings in real time. These issues often result in poor service accessibility, delayed transactions, and reduced customer satisfaction, especially in situations where customers need quick and convenient access to rental services.

To address these problems, the researchers designed a web-based reservation system that allows customers to browse available vehicles, view rental details, and submit booking requests through an online platform. The system also provides administrative functions that enable rental operators to manage vehicle listings, monitor reservations, and update booking statuses efficiently. By centralizing these operations into a single digital system, the platform improves coordination between customers and service providers while ensuring that rental data is properly organized and easily accessible.

The study found that implementing the system significantly improved reservation efficiency by reducing manual workload and minimizing delays in processing customer requests. It also enhanced data accuracy by reducing errors commonly associated with manual record-keeping methods. Furthermore, the system increased customer convenience by allowing users to access rental services remotely at any time, eliminating the need for physical visits or direct communication with staff for basic transactions.

The researchers concluded that web-based reservation systems are essential tools for modernizing car rental operations and improving overall business performance. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how integrating online booking capabilities and centralized data management can enhance operational efficiency, improve service delivery, and support accurate transaction processing.

ISSN : 2773-5141

Source: https://periodical.uthm.edu.my/index.php/aitcs/article/view/12321

Jadhav and Karande (2023) developed an online car rental system designed to improve the accessibility, convenience, and efficiency of vehicle rental services through a web-based platform. The researchers identified that traditional car rental processes often require customers to physically visit rental offices or communicate through phone calls to inquire about vehicle availability and complete reservations. These manual methods are time-consuming and prone to inefficiencies such as scheduling conflicts, delayed responses, and inaccurate record keeping, which can negatively affect both operational performance and customer satisfaction.

To address these issues, the researchers designed a web-based system that allows users to register accounts, browse available vehicles, and book rental services online. The platform provides a user-friendly interface that simplifies the reservation process and enables customers to complete transactions without the need for direct interaction with staff. On the administrative side, the system supports the management of vehicle inventories, booking records, and customer information, ensuring that all rental operations are properly organized and updated in real time.

The study found that the implementation of the online car rental system significantly improved booking efficiency by reducing the time required to process reservations and eliminating unnecessary manual procedures. It also enhanced data accuracy by centralizing records within a database, thereby minimizing errors and duplication of information. In addition, the system improved customer experience by providing faster access to rental services and real-time availability of vehicles, making the entire process more convenient and reliable.

The researchers concluded that web-based rental systems are effective solutions for modernizing traditional car rental operations and improving service delivery. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how integrating online booking functionality and centralized data management can enhance operational efficiency and ensure more accurate transaction and rental record processing.

ISSN: 2456-4184

Source: https://www.ijnrd.org/papers/IJNRD2304702.pdf

Bilal and Hanif (2025) developed a web-based vehicle rental application for **PT. Duta Rental Mobil** using the Waterfall software development methodology to improve the efficiency and accuracy of rental operations. The researchers identified that the company relied on manual processes for managing rental records, vehicle inventory, and transaction reports. These conventional practices often resulted in operational inefficiencies, inaccurate data, delayed transaction processing, and an increased risk of human error. As the number of customers grew, the manual system became more difficult to manage, making it challenging to organize rental schedules, monitor vehicle availability, and maintain accurate records.

To address these challenges, the researchers designed and implemented a web-based vehicle rental system using **PHP** and **MySQL**, following the Waterfall development model. The system provides features for customer registration, login, vehicle selection, online booking, payment processing, and rental history. It also includes administrative functions for managing customer information, vehicle data, reservations, and transaction records. By centralizing these operations into a computerized database, the application enables administrators to monitor rental activities more efficiently while providing customers with a more convenient reservation process.

The findings of the study showed that the developed application successfully improved operational efficiency by reducing manual data processing, minimizing recording errors, and accelerating rental transactions. System evaluation through alpha and beta testing demonstrated that the application functioned reliably, with beta testing achieving an **88.00%** user satisfaction score, indicating that users found the system functional, easy to use, and effective for managing rental operations. The researchers also noted that although the system met its objectives, future improvements such as integrating GPS-based vehicle tracking could further enhance its functionality.

The researchers concluded that implementing a web-based vehicle rental application using the Waterfall methodology significantly improves the management of rental transactions, customer information, and vehicle records while enhancing service quality and operational performance. This study is relevant to the proposed **Car Rental with Cash Register and Reservation System** because it demonstrates how a centralized web-based platform can automate reservation management, improve transaction accuracy, and streamline rental operations through an integrated information system.

ISSN: 2723-6927

Source: https://ejurnal.seminar-id.com/index.php/josh/article/view/7890

Ghiffari, Marwa, and Setiawan (2022) developed a web-based car rental reservation information system for rental businesses in Kampar Regency to improve the efficiency and accessibility of vehicle reservation services. The researchers identified that many car rental businesses still relied on conventional reservation methods, requiring customers to visit rental offices directly or contact operators through manual communication. These traditional processes limited customer access to rental information, increased the time required to complete reservations, and made it difficult for administrators to manage booking records and vehicle availability efficiently.

To address these challenges, the researchers designed and implemented a web-based reservation information system using **PHP** as the programming language and **MySQL** as the database management system. The system was developed following the **Waterfall** software development methodology, which includes the stages of planning, analysis, design, implementation, testing, and maintenance. The application enables customers to view available vehicles, access rental information, and submit reservations online without geographical or time limitations. It also provides administrators with features for managing customer records, monitoring reservations, maintaining vehicle data, and organizing rental transactions through a centralized database.

The findings of the study showed that the web-based reservation system successfully improved the efficiency of the rental process by reducing dependence on manual reservation methods and providing customers with easier access to rental services. The centralized database improved the accuracy of customer, vehicle, and reservation records while simplifying administrative tasks related to booking management. The researchers also emphasized that the online platform allowed customers to obtain information and complete reservations anytime and anywhere, thereby improving service accessibility and supporting more organized rental operations.

The researchers concluded that implementing a web-based reservation information system is an effective solution for modernizing car rental services by improving reservation efficiency, data management, and customer convenience. This study is relevant to the proposed **Car Rental with Cash Register and Reservation System** because it demonstrates how integrating an online reservation platform with centralized data management can improve operational efficiency, streamline rental transactions, and provide more reliable customer service.

ISSN: 2809-0446

Source: https://j-innovative.org/index.php/Innovative/article/view/71

Irwanto, Yulianti, Solehudin, and Voutama (2022) designed a web-based vehicle rental application to improve the efficiency of rental services and simplify the booking process through digital technology. The researchers identified that many vehicle rental businesses still experience difficulties in managing customer orders, promoting available vehicles, and organizing rental information due to the continued use of conventional methods. Manual reservation processes often require direct communication between customers and rental operators, which can slow down transactions, limit service accessibility, and increase the possibility of errors in recording rental data. These challenges affect the ability of rental businesses to provide fast and reliable services to customers.

To address these issues, the researchers developed a website-based rental application using **PHP** and **MySQL** to support online vehicle booking and information management. The development process involved requirements analysis, system design, database design, and interface design. The system provides customers with access to vehicle information and online ordering features, while administrators can manage rental data more efficiently through a centralized database. By integrating vehicle records, customer information, and booking transactions into a single platform, the application improves data organization and supports smoother rental operations.

The findings of the study showed that the developed web-based rental application successfully improved the management of rental activities by making the booking process faster and more organized. The system reduced the need for paper-based invoices and manual recording, allowing rental businesses to manage transactions more efficiently. In addition, the online platform helped improve service accessibility by allowing customers to obtain rental information and submit orders through a website. The researchers also emphasized that the application can serve as a promotional medium for rental businesses by providing customers with easier access to available services.

The researchers concluded that implementing a web-based vehicle rental application using PHP and MySQL is an effective approach for improving rental business operations, enhancing booking efficiency, and reducing errors caused by manual processes. This study is relevant to the proposed **Car Rental with Cash Register and Reservation System** because it demonstrates how web-based automation and centralized data management can improve reservation handling, organize transaction records, and support more efficient vehicle rental operations.

ISSN: P-ISSN: 1907-0012 E-ISSN: 2714-5417

Source: https://journal.stekom.ac.id/index.php/elkom/article/view/621

Wetapo and Trisnawati (2024) developed a web-based car rental application for **Adiba Trans Rental** using **PHP** and **MySQL** to improve the efficiency of rental management and provide a more organized platform for handling vehicle rental activities. The researchers identified that the increasing demand for car rental services created challenges for rental businesses in managing customer information, vehicle availability, reservation processes, and transaction records. Traditional rental methods often make it difficult for businesses to efficiently monitor rental activities, maintain accurate records, and provide convenient services to customers. These limitations may result in slower transaction processing and reduced effectiveness in managing rental operations.

To address these issues, the researchers designed and implemented a web-based car rental application that integrates customer management, vehicle management, reservation processing, and payment recording into a centralized system. The application was developed using **PHP** as the programming language and **MySQL** as the database management system. The system allows customers to register accounts, log in, view available vehicles, submit rental requests, and provide payment information through an online platform. Meanwhile, administrators can manage user data, vehicle records, rental transactions, and generate rental reports through the system. By combining these functions into one platform, the application improves data organization and supports more efficient rental operations.

The findings of the study showed that the developed web-based rental application successfully achieved its objectives by providing complete functional features, an easy-to-use interface, and effective integration between PHP and MySQL. The system helped simplify rental management processes by reducing dependence on manual procedures and improving the organization of customer, vehicle, and transaction data. The researchers also emphasized that the application provides an effective solution for managing car rental services through a web platform, allowing users to access rental services more conveniently while helping administrators handle rental operations more efficiently.

The researchers concluded that implementing a web-based car rental application is an effective approach for improving rental service management, increasing operational efficiency, and enhancing customer convenience. This study is relevant to the proposed **Car Rental with Cash Register and Reservation System** because it demonstrates how integrating web technologies, centralized databases, and automated rental processes can improve reservation handling, maintain accurate transaction records, and support more efficient vehicle rental operations.

ISSN: P-ISSN: 3031-9935 E-ISSN: 3031-9943

Source: https://doi.org/10.61132/saturnus.v2i4.348

Soares, da Costa, and Bernardo (2025) developed a web-based car rental management application system for **Kailelo Car Rental** using **PHP** and **MySQL** to improve the efficiency of vehicle rental operations and enhance customer service. The researchers identified that the rental company relied on manual methods for managing vehicle data, customer records, and rental transactions, resulting in inefficient data processing, difficulties in organizing rental information, and delays in serving customers. These conventional practices also increased the risk of data inaccuracies and made it challenging for administrators to monitor rental activities effectively.

To address these challenges, the researchers designed and implemented a web-based management system using the **Waterfall** software development methodology. The system was developed with **PHP** as the programming language and **MySQL** as the database server, while **UML** and **Entity Relationship Diagrams (ERD)** were used during system design. The application provides several administrative modules, including a dashboard, brand management, vehicle data management, customer records, destination management, order processing, account management, and report generation for customers, rentals, and vehicles. By centralizing these functions into one platform, the system enables administrators to manage rental operations more efficiently while maintaining organized and accurate records.

The findings of the study revealed that the implementation of the web-based management application successfully improved the efficiency of rental operations by simplifying data management and accelerating transaction processing. The researchers reported that all system modules functioned properly, allowing new customers to receive faster and more efficient services while enabling rental business owners to manage operational data more effectively. The computerized system also improved the organization of rental records and supported better monitoring of business activities compared to manual procedures.

The researchers concluded that implementing a web-based car rental management application using PHP and MySQL is an effective solution for modernizing rental business operations, improving customer service, and enhancing data management efficiency. This study is relevant to the proposed **Car Rental with Cash Register and Reservation System** because it demonstrates how integrating web-based reservation and management features with a centralized database can streamline rental transactions, improve record accuracy, and support more efficient business operations.

Source: https://dilijetin.org/index.php/dilijetin/article/view/1

Waje et al. (2024), in their study titled *"Car Rental System,"* developed a computerized car rental management system to improve the efficiency of vehicle rental operations and provide customers with a faster and more convenient reservation experience. The researchers identified that many car rental businesses continue to depend on manual procedures for handling reservations, maintaining customer records, tracking vehicle availability, and processing rental transactions. These conventional methods often lead to booking conflicts, inaccurate record-keeping, delayed transaction processing, and difficulties in monitoring the status of rented and available vehicles. Such operational challenges reduce service efficiency, increase the likelihood of human error, and negatively affect customer satisfaction.

To address these issues, the researchers designed and implemented a digital car rental system that automates the core functions of the rental business, including customer registration, vehicle inventory management, reservation processing, billing, and rental record management. The system enables customers to browse available vehicles, submit rental requests, and complete booking transactions through a user-friendly interface. At the same time, administrators can efficiently manage customer information, monitor vehicle availability, process reservations, and maintain accurate transaction records using a centralized database. By integrating these functions into a single platform, the system streamlines rental operations and reduces the dependence on manual documentation.

The findings of the study showed that the developed car rental system significantly improved the efficiency and accuracy of rental business operations. The automated reservation and record management features minimized booking errors, reduced the time required to process customer transactions, and improved the organization of rental information. Furthermore, the centralized database allowed administrators to monitor vehicle availability in real time and generate accurate records for business operations. The researchers also noted that the system enhanced customer convenience by providing a more accessible and reliable reservation process while improving the overall quality of service offered by the rental company.

The researchers concluded that implementing a computerized car rental management system is an effective solution for modernizing vehicle rental operations, improving reservation efficiency, and maintaining accurate customer and transaction records. This study is relevant to the proposed **Car Rental with Cash Register and Reservation System** because it supports the use of digital automation to streamline reservation processing, organize rental and payment records, improve vehicle management, and enhance the overall efficiency and reliability of car rental business operations.

ISSN: 2583-7877

Source: https://doi.org/10.63671/ijsssr.v2i1.240

Nahnisha et al. (2022), in their study titled *"Online Car Rental System,"* developed a web-based car rental system to improve the efficiency of vehicle rental services and provide customers with a more convenient and accessible reservation process. The researchers identified that many car rental businesses still rely on traditional methods of handling reservations and customer transactions, which often require customers to visit the rental office or communicate directly with staff before booking a vehicle. These manual procedures consume considerable time, limit customer convenience, and increase the possibility of errors in recording rental information and managing vehicle availability. As a result, rental companies experience operational inefficiencies that affect both customer satisfaction and business performance.

To address these issues, the researchers designed and implemented an online car rental system using web technologies supported by a MySQL database and the XAMPP server environment. The development process focused on automating essential rental functions, including customer registration, vehicle selection, online reservation, and customer information management. The system allows customers to browse available vehicles, submit booking requests, and specify their rental requirements from any location through an internet connection. Meanwhile, administrators can efficiently maintain customer records, monitor vehicle availability, and manage rental transactions through a centralized digital platform. By integrating these functions into a single system, the application reduces dependence on manual processes and improves the organization of rental information.

The findings of the study showed that the developed online car rental system successfully enhanced the efficiency of rental operations by simplifying reservation procedures and improving the accuracy of customer and vehicle records. The web-based platform enabled customers to make reservations more conveniently while providing rental companies with an organized method of managing transactions and monitoring available vehicles. The researchers also emphasized that automating rental services through an online platform improves business productivity, minimizes human errors, and delivers faster and more reliable customer service.

The researchers concluded that implementing an online car rental system is an effective approach to modernizing vehicle rental operations by improving reservation efficiency, enhancing data management, and providing customers with greater accessibility to rental services. This study is relevant to the proposed **Car Rental with Cash Register and Reservation System** because it demonstrates how web-based automation can streamline reservation handling, maintain accurate customer and vehicle records, improve operational efficiency, and provide a more convenient and reliable rental experience for both customers and business administrators.

ISSN: 2320-088X

Source: https://ijcsmc.com/docs/papers/March2022/V11I3202226.pdf

Maulana et al. (2024), in their study titled *"Application of the Rapid Application Development (RAD) Method in Designing a Web-Based Car Rental Information System,"* developed a web-based car rental information system using the Rapid Application Development (RAD) methodology to improve the efficiency of vehicle rental operations. The researchers identified that many car rental businesses still rely on manual procedures for recording customer information, processing rental transactions, and preparing rental reports. These conventional practices often result in inefficient data management, delayed transaction processing, inaccurate record-keeping, and difficulties in generating reports. Such challenges reduce operational efficiency and make it difficult for rental companies to provide fast and reliable services to customers.

To address these issues, the researchers designed and implemented a web-based car rental information system following the Rapid Application Development (RAD) approach. The development process emphasized rapid prototyping, continuous user involvement, and iterative system improvement to ensure that the application met the operational requirements of the rental business. The system computerized essential rental functions, including customer registration, vehicle management, rental transaction processing, and report generation. Through a centralized database, administrators can efficiently manage customer records, monitor vehicle availability, and process rental transactions, while customers benefit from a more organized and convenient rental process.

The findings of the study showed that the developed web-based information system successfully improved the management of car rental operations by automating business processes that were previously handled manually. The system reduced errors in recording rental transactions, simplified report generation, and increased the efficiency of managing customer and vehicle information. The researchers also found that applying the RAD methodology accelerated the software development process while producing a system that effectively addressed the operational needs of the rental company. As a result, both business administrators and customers experienced more efficient and reliable rental services.

The researchers concluded that implementing a web-based car rental information system using the Rapid Application Development (RAD) method is an effective solution for modernizing rental business operations, improving transaction efficiency, and minimizing errors associated with manual processes. This study is relevant to the proposed **Car Rental with Cash Register and Reservation System** because it demonstrates how rapid system development and web-based automation can improve reservation management, organize customer and transaction records, streamline business operations, and enhance the overall quality of vehicle rental services.

ISSN: 2597-3673

Source: https://journal.stmikjayakarta.ac.id/index.php/jisicom/article/view/1515

Qi, Shibghatullah, and Subaramaniam (2024), in their study titled *"Rehabilitating Flood-Damaged Cars for Sustainable Car Rental Services: A Web-Based System,"* developed a web-based car rental system that promotes sustainability by utilizing properly rehabilitated flood-damaged vehicles for rental services. The researchers identified that environmental concerns, increasing transportation costs, and the growing demand for affordable mobility solutions have encouraged the exploration of more sustainable approaches in the car rental industry. They also noted that many flood-damaged vehicles are discarded despite having the potential to be restored safely for continued use, resulting in unnecessary waste and environmental impact. At the same time, conventional rental services often lack digital platforms that effectively support sustainable vehicle management and transparent customer access to vehicle information.

To address these issues, the researchers designed and implemented a web-based car rental system using the Rapid Application Development (RAD) methodology. The development process involved requirement planning, user-centered design, rapid construction, and system implementation to produce a functional and efficient rental platform within a short development period. The system enables customers to browse available rehabilitated vehicles, make online reservations, and access rental services through a user-friendly interface. Administrators can efficiently manage vehicle records, rental transactions, and customer information using a centralized database. The researchers also proposed future enhancements such as GPS tracking and transparent disclosure of each vehicle's rehabilitation history to strengthen customer trust and improve service reliability.

The findings of the study showed that the developed web-based system successfully met its objectives by providing an efficient and accessible platform for sustainable car rental services. User acceptability testing indicated positive feedback regarding the system's usability, efficiency, and overall functionality. The researchers found that customers were willing to consider renting rehabilitated flood-damaged vehicles provided that the vehicles were properly repaired, safe to operate, and supported by transparent information regarding their condition. The study also demonstrated that integrating sustainability with digital technology can reduce environmental waste while providing cost-effective transportation alternatives.

The researchers concluded that implementing a web-based car rental system for rehabilitated flood-damaged vehicles is an effective approach to promoting sustainable transportation, improving rental service efficiency, and reducing vehicle waste through digital innovation. This study is relevant to the proposed **Car Rental with Cash Register and**

Reservation System because it demonstrates how web-based automation, centralized data management, and user-friendly reservation features can improve rental operations while supporting transparency, operational efficiency, and sustainable business practices.

ISSN: 2188-7829

Source: https://alife-robotics.co.jp/LP/2024/GS7-1.htm

These foreign studies collectively demonstrate that modern car rental systems significantly benefit from the integration of automation, real-time communication, data-driven decision-making, and secure transaction technologies. Across the reviewed literature, it is evident that the shift from manual to web-based systems enhances operational efficiency by reducing human intervention in reservation handling, payment processing, and record management. In addition, the incorporation of centralized databases allows for more accurate data storage and faster retrieval of information, which improves both administrative performance and customer service delivery. The studies also emphasize that real-time communication features, such as notifications and online updates, play a crucial role in increasing transparency and strengthening customer trust in rental services.

Furthermore, several studies highlight the importance of advanced technologies such as analytics, blockchain, and structured system development methodologies in improving system reliability, scalability, and security. These technologies enable rental companies to optimize fleet management, ensure transaction integrity, and make informed business decisions based on accurate data insights. As a result, organizations are able to reduce operational costs while improving resource allocation and service availability.

Overall, the reviewed literature suggests that integrating multiple system components such as reservation management, payment processing, communication modules, and data analytics creates a more efficient and user-friendly car rental platform. This integration not only streamlines internal operations but also enhances customer experience by providing faster, more reliable, and more accessible services. These findings strongly support the development of the proposed Car Rental with Cash Register System, as they emphasize the need for a unified, automated, and data-driven platform that can effectively manage both rental operations and financial transactions while maintaining accuracy, security, and efficiency.

2.2 Local Studies

The continuous development of information technology in the Philippines has significantly influenced the improvement and modernization of various service-oriented industries, including transportation and vehicle rental services. As the demand for faster, more efficient, and more accessible services continues to grow, many local systems have gradually shifted from traditional manual processes to computerized and web-based platforms. This transition is driven by the need to reduce operational inefficiencies, minimize human errors, and improve the overall quality of service delivery.

In many local settings, especially within small to medium-sized rental businesses, traditional car rental operations still rely on manual record-keeping, walk-in transactions, and basic communication methods such as phone calls or messaging applications. These processes often lead to challenges such as booking conflicts, delayed transaction processing, difficulty in tracking vehicle availability, and inconsistencies in customer records. As a result, service providers experience reduced efficiency, while customers encounter inconvenience and longer waiting times.

To address these issues, several local studies in the Philippines have explored the development and implementation of web-based systems that automate reservation processes, centralize data management, and improve transaction handling. These systems are designed to provide real-time access to information, allowing both administrators and users to efficiently manage bookings, monitor availability, and process transactions more accurately.

Furthermore, local literature emphasizes that the integration of automated systems not only improves operational performance but also enhances customer satisfaction by providing more convenient and accessible services. The use of centralized databases, user-friendly interfaces, and structured system development methodologies has been shown to significantly improve system reliability and efficiency.

Overall, local studies in the Philippines consistently demonstrate that web-based systems are essential in modernizing rental and service-based operations. They highlight the importance of automation and digital transformation in improving efficiency, accuracy, and accessibility, which strongly supports the development of a Car Rental with Cash Register System.

Albino (2021) developed a car rental management system with a scheduling algorithm aimed at improving the efficiency, accuracy, and reliability of rental business operations. The study identified that traditional car rental systems often rely on manual processes for managing vehicle reservations, scheduling, and inventory tracking. These manual methods frequently result in inefficiencies such as booking conflicts, delayed transaction processing, and difficulties in monitoring vehicle availability, which can negatively affect overall service quality and customer satisfaction.

To address these challenges, the researcher developed a computerized car rental management system that integrates a scheduling algorithm to automate reservation handling and vehicle allocation. The system was designed to manage key functions such as booking management, vehicle inventory tracking, customer transactions, and scheduling of rentals. By incorporating algorithm-based scheduling, the system ensures that vehicle assignments are optimized and conflicts in reservations are minimized.

The findings of the study revealed that the implemented system improved transaction speed, reduced manual workload, and enhanced the accuracy of rental operations. It also demonstrated that automated scheduling significantly improves resource allocation and ensures more efficient utilization of available vehicles. Furthermore, the system received positive feedback in terms of usability and performance, indicating its effectiveness in addressing operational challenges in rental businesses.

The researcher concluded that integrating scheduling algorithms into car rental systems significantly improves operational efficiency and service reliability. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how automation and intelligent scheduling can enhance booking accuracy, improve transaction management, and optimize overall system performance.

ISSN: 2348-4098

Source:
https://www.researchgate.net/publication/350603452_Development_of_Car_Rental_Management_System_with_Scheduling_Algorithm

Golo and Encarnacion (2024) developed a study titled *"Revolutionizing the Rental Services in Siargao Island: Basis for Developing an Online Vehicle Rental Management System"*, which focused on addressing the inefficiencies and challenges experienced in the vehicle rental industry within Siargao Island, Philippines. The researchers identified that many local rental service providers still rely on manual processes such as walk-in transactions, paper-based records, and informal booking systems. These traditional methods often result in operational problems such as booking conflicts, difficulty in tracking vehicle availability, delayed confirmations, and inconsistent record management. Such issues negatively affect both service providers and customers, especially in a tourism-driven location where fast and reliable transportation services are essential.

To address these challenges, the study proposed the development of an online vehicle rental management system that would automate reservation processes and centralize rental operations into a digital platform. The system was designed to provide real-time updates on vehicle availability, streamline booking procedures, and improve communication between customers and rental operators. It also aimed to enhance fleet management by allowing administrators to efficiently monitor vehicle usage and status.

The findings of the study revealed that the implementation of a web-based rental system significantly improves operational efficiency by reducing manual workload and minimizing human errors in reservation handling. It also enhances accessibility by allowing customers to reserve vehicles anytime and anywhere without physically visiting rental offices. Furthermore, the system improves service reliability by ensuring that rental information is updated in real time and properly stored in a centralized database.

The researchers concluded that transitioning from manual to automated rental systems is essential for improving the efficiency and competitiveness of rental businesses in Siargao Island. This study is highly relevant to the proposed Car Rental with Cash Register System because it demonstrates the importance of integrating online reservation functionality, centralized databases, and automated transaction processing to improve both operational performance and customer satisfaction.

ISSN: 2581-9429 Source:
https://www.researchgate.net/publication/381175233_Revolutionizing_the_Rental_Services_in_Siargao_Island_Basis_for_Developing_an_Online_Vehicle_Rental_Management_System

Anglo et al. (2022), published in the *2022 IEEE 14th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, developed a system titled *Book-Express* which is a web-based and mobile reservation and booking platform designed to improve the efficiency of UV Express transportation services in the Philippines. The researchers identified that traditional UV Express operations often experience significant challenges such as long queues at terminals, extended waiting times for passengers, and inefficient scheduling between drivers and commuters. These issues highlight the need for a more organized and technology-driven system that can streamline the booking and reservation process.

To address these problems, the study proposed a reservation system that allows passengers to book seats through a website and mobile application. The system aims to reduce congestion at terminals by enabling advance booking and improving the overall flow of passenger allocation. It also enhances communication between drivers and passengers by providing a structured and centralized platform for reservations. The system's performance was evaluated through user acceptance testing based on criteria such as functionality, usability, reliability, performance, and supportability.

The findings of the study revealed that the developed system achieved an overall rating of 4.60 (Excellent) for the mobile application and 4.53 (Excellent) for the website, indicating high user satisfaction and system effectiveness. The results demonstrate that the implementation of a digital reservation system significantly improves transportation efficiency, reduces waiting time, and enhances service convenience for users.

This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how web-based reservation platforms can improve operational efficiency, reduce manual transaction processes, and enhance user experience through automated booking and centralized system management.

The researchers concluded that web-based reservation systems play a crucial role in improving service delivery and operational efficiency. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how automated booking management and centralized data processing can enhance transaction speed, improve accuracy, and support more efficient service operations.

Conference ISBN: 978-1-6654-6494-9

Source: https://ieeexplore.ieee.org/document/10109523

Grepon (2022) developed a study titled *Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines*, which focused on improving the efficiency of institutional data management through a web-based information system. The researcher identified that many educational institutions still relied on manual and fragmented methods of managing records, resulting in delays in processing information, data inconsistencies, and difficulties in maintaining accurate and up-to-date records. These challenges negatively affected administrative efficiency and decision-making processes.

To address these issues, the study employed the **Agile software development methodology** in designing and implementing a web-based school management information system. The system centralized institutional data into a single database and provided features such as student information management, record maintenance, report generation, and user account management. By automating routine administrative processes, the system improved the accessibility, organization, and reliability of institutional records while allowing authorized users to retrieve and update information efficiently.

The findings of the study revealed that the developed system significantly improved the efficiency and accuracy of information management by reducing manual processing, minimizing data redundancy, and enhancing the overall usability of the system. User evaluation results indicated that the web-based system was effective in supporting daily administrative operations and improving the management of institutional records. The study also demonstrated that applying the Agile development methodology contributed to the successful implementation of a reliable and user-friendly information system.

The researcher concluded that implementing a centralized web-based information system using a structured software development methodology enhances operational efficiency, data accuracy, and overall system reliability. This study is relevant to the proposed **Car Rental with Cash Register System** because it demonstrates the importance of centralized database management, automated information processing, and systematic software development in building efficient systems that improve transaction processing, record management, and operational performance.

ISSN: 2546-115X

Source: https://doi.org/10.25147/ijcsr.2017.001.1.74

Marafo et al. (2025), in their study titled *"Arkila: A Progressive Web App Car Rental Management System for Kataguan Rides in La Trinidad, Benguet,"* developed a progressive web-based car rental management system aimed at improving the efficiency and accessibility of rental services. The researchers identified that many local car rental operations still rely on traditional manual processes such as phone calls, text messaging, and handwritten records, which often result in scheduling conflicts, delayed confirmations, and inefficient communication between customers and service providers. These issues contribute to reduced operational efficiency and lower customer satisfaction.

To address these challenges, Marafo et al. (2025) developed a progressive web application that allows users to browse available vehicles, make reservations online, and manage booking transactions through a centralized platform. The system was designed using Feature-Driven Development (FDD), ensuring that system features are developed incrementally based on user requirements. It also utilizes modern web technologies to improve responsiveness, scalability, and overall system performance.

The findings of the study revealed that the implementation of the system significantly improved booking efficiency by reducing reliance on manual communication and streamlining reservation processes. It also enhanced operational management by providing a centralized system for monitoring vehicle availability and managing bookings in real time. Furthermore, user evaluation results indicated a high level of satisfaction, showing that the system effectively addresses common inefficiencies in traditional rental operations.

The researchers concluded that web-based and progressive web applications are essential tools for modernizing car rental services by improving accessibility, efficiency, and reliability. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how digital systems can enhance reservation management, improve operational workflows, and support better customer service in the car rental industry.

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Source: https://sajst.pti.edu.ph/online/index.php/sajst/issue/view/12

Estoque et al. (2026), in their study titled *"The Role of Assurance and Safety in Shaping Customer Referrals: Insights from Car Rental Services in Malaybalay City,"* examined how assurance and safety factors influence customer satisfaction and referral behavior within car rental services in the Philippines. The researchers identified that service quality in car rental operations is strongly affected by customer perception of safety, trust, and reliability throughout the entire rental process, from booking to vehicle return. These factors are especially important in ensuring repeat usage and positive word-of-mouth recommendations.

To address these concerns, the study analyzed customer experiences and service delivery practices within car rental businesses in Malaybalay City. The researchers emphasized that ensuring safety measures, transparent transactions, and reliable service procedures significantly improves customer confidence. They also highlighted that service assurance plays a critical role in connecting service quality with overall customer satisfaction, particularly in the context of Philippine transportation services.

The findings of the study revealed that higher levels of assurance and safety strongly contribute to increased customer satisfaction and a greater likelihood of customer referrals. It was also observed that customers are more likely to trust and repeatedly use car rental services that demonstrate consistent reliability and secure transaction processes. Furthermore, the study highlighted that improving service quality directly enhances business reputation and operational success.

The researchers concluded that assurance and safety are essential components in shaping customer behavior and improving service outcomes in car rental systems. This study is relevant to the proposed Car Rental with Cash Register System because it highlights the importance of secure transactions, reliable booking processes, and customer trust in improving overall system performance and business success.

ISSN: 3107-7854

Source: https://ijsair.com/index.php/ijsair/article/view/51

Balmes et al. (2022) developed a study titled *"PARADA: Parking Space Rental and Leasing Application System,"* which focused on addressing the increasing demand for parking spaces in urban areas in the Philippines due to the continuous rise of vehicle ownership. The researchers identified that limited parking availability, especially in densely populated locations, has become a significant problem for drivers, leading to inconvenience, inefficiency, and wasted time searching for available parking spaces. This issue highlights the need for a digital solution that can efficiently connect users to available parking resources in real time.

To address these challenges, the researchers developed PARADA, a mobile-based parking application system that allows users to locate and reserve available parking spaces based on their current location. The system utilizes privately owned parking spaces from landholders in establishments such as subdivisions and condominiums, thereby maximizing the use of existing parking resources. The system was developed using the Scrum methodology, which allowed iterative development and continuous improvement based on user feedback.

The findings of the study revealed that the implementation of the PARADA system significantly improved accessibility and convenience for users by enabling real-time discovery of available parking spaces. It also enhanced operational efficiency by optimizing the utilization of private parking facilities and reducing the time spent by drivers searching for parking. Furthermore, user evaluation results indicated positive feedback in terms of system usability, safety, functionality, and overall user experience.

The researchers concluded that mobile-based parking management systems are effective solutions for addressing urban parking shortages and improving transportation efficiency. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how digital platforms can improve resource management, enhance user convenience, and streamline service operations through automation and real-time system functionality.

ISBN (Conference Proceedings): 978-1-6654-4434-7 (IEEE Conference Proceedings)

Source: https://ieeexplore.ieee.org/document/9720838

Abat et al. (2023), in their study titled *"Online Car Rental System,"* developed a web-based car rental platform aimed at improving the efficiency, accessibility, and overall management of rental services. The researchers identified that many traditional car rental operations still rely heavily on manual booking procedures, walk-in transactions, and fragmented record-keeping systems. These conventional methods often result in several operational problems such as delayed booking confirmations, inconsistencies in transaction records, difficulties in monitoring vehicle availability, and inefficient handling of customer requests. As a result, these issues negatively affect both service providers and customers by reducing operational speed and service quality.

To address these challenges, the researchers developed a centralized web-based system that integrates essential car rental functionalities into a single platform. The system allows users to conveniently browse available vehicles, view rental details, and make online reservations without the need for physical visits or manual coordination. On the administrative side, the system enables efficient management of bookings, monitoring of vehicle availability, and processing of rental transactions in real time. By centralizing data and automating key processes, the system reduces redundancy, minimizes human errors, and improves overall workflow efficiency.

The findings of the study revealed that the implementation of the system significantly improved usability, reliability, and operational efficiency within car rental services. It also enhanced data accuracy and reduced delays in processing transactions, leading to better coordination between customers and service providers. Furthermore, the system demonstrated improved service delivery by providing users with faster access to information and more convenient reservation processes.

The researchers concluded that web-based car rental systems play a crucial role in modernizing rental operations by improving efficiency, accuracy, and customer satisfaction. This study is relevant to the proposed Car Rental with Cash Register System because it demonstrates how centralized booking systems and automated transaction handling can enhance operational performance and improve the overall management of rental services.

Source: https://lakasa.dmmmsu.edu.ph/entities/publication/72145226-23ba-4dd0-ab21-2d1bca1276cf/full

Balintag et al. (2025), in their study titled *"SORMS: Steffi's Online Reservation Management System,"* developed a web-based reservation management system designed to improve the efficiency, accuracy, and reliability of reservation processes at Steffi's Garden Resort. The researchers identified that the resort's existing manual reservation process resulted in operational inefficiencies, including booking conflicts, delayed reservation confirmations, inaccurate record-keeping, and difficulties in managing customer information and payment transactions. These challenges negatively affected both customer satisfaction and the overall productivity of the resort's operations.

To address these challenges, the researchers developed the Steffi's Online Reservation Management System (SORMS) using the Agile Scrum software development methodology. The system integrates online reservation, real-time booking management, digital payment processing, and centralized customer and reservation records into a single web-based platform. It enables customers to conveniently check availability and make reservations online, while administrators can efficiently monitor bookings, manage customer information, update reservation schedules, and generate accurate records. The centralized system minimizes manual intervention, reduces booking errors, and streamlines the entire reservation workflow.

The findings of the study revealed that the implemented system significantly enhanced reservation management by reducing booking errors by approximately 94% and improving reservation processing speed by around 80% compared to the previous manual process. The system also received excellent evaluation ratings based on the ISO 25010 software quality model and the Technology Acceptance Model (TAM), demonstrating high levels of functional suitability, usability, reliability, and user acceptance. These results indicate that the system effectively improved operational efficiency while providing a more convenient reservation experience for both customers and administrators.

The researchers concluded that implementing a web-based reservation management system is an effective approach to modernizing reservation operations by improving efficiency, accuracy, and customer satisfaction. This study is relevant to the proposed **Car Rental with**

Cash Register System because it demonstrates how centralized reservation management, real-time transaction processing, and automated record management can enhance operational performance and service quality. The concepts and technologies presented in the study support the development of an integrated reservation and transaction system capable of reducing manual errors and improving business operations in the car rental industry.

ISSN: 2822-4353

Source: https://www.ejournals.ph/article.php?id=32526

Dagami et al. (2024), in their study titled *"School Management System: Logistics Management System (Project Management, Warehousing, Procurement, Vendor Portal, Audit Management, Vehicle Reservation System, Document Tracking System),"* developed a web-based logistics management system designed to improve the efficiency, organization, and coordination of logistics operations within an educational institution. The researchers identified that the school's existing logistics processes relied heavily on manual procedures handled by the property custodian, resulting in inefficient procurement, inventory management, item distribution, and vehicle reservation. These manual practices often led to delays, increased workload, inaccurate record-keeping, and difficulties in monitoring logistics-related transactions.

To address these challenges, the researchers developed an integrated logistics management system using the Agile software development methodology. The system incorporates several modules, including project management, warehousing, procurement, vendor portal, audit management, vehicle reservation, and document tracking. These modules enable administrators to manage inventory, monitor procurement requests, reserve vehicles, track documents, and oversee logistics operations through a centralized platform. By automating these processes, the system minimizes manual intervention, improves data accuracy, and streamlines overall logistics management.

The findings of the study revealed that the developed system enhanced operational efficiency by improving stock handling, procurement management, and logistics coordination. The implementation of the integrated modules reduced administrative workload, facilitated more organized inventory monitoring, and enabled more efficient management of vehicle reservations and document tracking. The researchers also found that the Agile development approach allowed the system to meet its intended functionality while providing opportunities for future enhancements based on user feedback.

The researchers concluded that implementing an integrated logistics management system significantly improves the efficiency, accuracy, and reliability of logistics operations within organizations. This study is relevant to the proposed **Car Rental with Cash Register System** because it demonstrates how centralized management, vehicle reservation, automated record-keeping, and integrated transaction processing can improve operational performance and reduce errors. The concepts presented in the study support the development of an efficient reservation and management system for car rental businesses.

ISSN: 2661-4472

Source: https://ojs.aaresearchindex.com/index.php/aasgbcpjmra/article/view/14210

Dalisay (2019), in the study titled *"Development of a Multi-platform Online Reservation System for Car Services,"* developed a multi-platform online reservation system designed to improve the efficiency, accessibility, and management of car service appointments. The researcher identified that many automotive service providers still relied on manual appointment scheduling, walk-in transactions, and paper-based record-keeping, which often resulted in delayed appointment confirmations, scheduling conflicts, inaccurate customer records, and inefficient service management. These operational challenges affected both service providers and customers by reducing productivity and causing inconvenience during the reservation process.

To address these challenges, the researcher developed a web- and mobile-based reservation system that allows customers to schedule car service appointments anytime and anywhere using multiple platforms. The system enables users to create reservations, view available service schedules, receive appointment updates, and monitor the status of their bookings. On the administrative side, the system provides centralized management of customer information, appointment schedules, and service records, allowing staff to efficiently monitor reservations and organize daily operations. By automating these processes, the system minimizes manual errors, improves data accuracy, and streamlines the overall reservation workflow.

The findings of the study revealed that the implementation of the multi-platform reservation system significantly improved the efficiency and reliability of appointment scheduling. The system reduced scheduling conflicts, accelerated reservation processing, enhanced record management, and provided customers with a more convenient and accessible method of booking car services. The evaluation also indicated that users found the system functional, user-friendly, and effective in supporting the operational requirements of automotive service businesses.

The researcher concluded that implementing a multi-platform online reservation system is an effective solution for modernizing car service operations by improving efficiency, accuracy, and customer satisfaction. This study is relevant to the proposed **Car Rental with Cash Register System** because it demonstrates how automated reservation management, centralized record-keeping, and digital transaction processing can enhance operational performance, reduce manual errors, and provide faster, more reliable services for customers. The concepts presented in the study support the development of an integrated reservation and cash register system for car rental businesses.

ISSN: 2651-6691

Source: https://www.studocu.com/ph/document/bulacan-state-university/bachelor-of-science-and-information-technology/397-article-text-1101-1-10-20220309/50832654

Libunao et al. (2025), in their study titled **"TrialaTS: Triala Transportation System,"** developed an online transportation management system designed to improve the accessibility, efficiency, and reliability of transportation services in Barangay Triala, Guimba, Nueva Ecija, Philippines. The researchers identified that traditional transportation operations often depend on manual processes, which may cause difficulties in managing reservations, monitoring vehicle availability, and providing convenient access to transportation services. These limitations can result in longer waiting times, inefficient scheduling, and challenges in organizing transportation transactions between passengers and service providers.

To address these concerns, the researchers developed TrialaTS, a web-based transportation system that allows users to access transportation services through an online platform. The system provided features such as user registration, transportation booking, reservation management, and administrative monitoring to improve the overall organization of transportation operations. By digitizing the reservation process, the system aimed to minimize manual work, reduce transaction delays, and provide users with a more convenient method of requesting transportation services.

The researchers evaluated the system using the ISO 25010 software quality model, focusing on important software characteristics such as functional suitability, usability, security, and reliability. The findings showed that the developed system was capable of improving transportation service management by providing a structured and efficient approach to handling reservations and user transactions. The study emphasized that implementing computerized transportation systems can help organizations improve operational efficiency, enhance customer convenience, and reduce issues commonly experienced in manual-based transportation management.

ISSN: 2822-4353

Source: https://doi.org/10.70838/pemj.370507

Abadilla et al. (2015), in their study titled **"Online Bus Reservation and Tracking System,"** developed an online-based transportation system designed to improve the efficiency of bus reservation processes and provide better monitoring of bus locations during travel. The researchers identified that many bus companies still rely on traditional methods, such as manual ticket purchasing and informal scheduling practices, which may cause difficulties in managing passenger reservations and determining the current location of buses. These issues can lead to delays, inconvenience for passengers, and challenges in providing reliable transportation services.

To address these concerns, the researchers developed an online bus reservation and tracking system that serves as an alternative to manual ticketing procedures. The system aimed to provide passengers with a more convenient way of reserving bus seats while allowing transportation providers to improve the management of their services. Through the integration of online reservation features and tracking capabilities, the system helped organize passenger information, reservation records, and bus location monitoring. The researchers conducted interviews with personnel from a selected bus company to understand existing operational processes and identify system requirements.

The study emphasized that implementing an online transportation management system can improve service accessibility, reduce manual transaction processes, and enhance the overall efficiency of bus operations. The researchers highlighted that internet-based systems provide opportunities for transportation companies to offer faster and more organized services to customers.

ISSN: 2244-6451

Source: https://www.ejournals.ph/article.php?id=12679

These local studies collectively demonstrate that the adoption of web-based systems in the Philippines plays a significant role in improving operational efficiency, accuracy, and overall service delivery across various industries, particularly in transportation and rental services. Most of the reviewed studies highlight that traditional manual processes—such as paper-based records, walk-in transactions, and fragmented booking systems—often lead to operational inefficiencies, including data inconsistencies, booking conflicts, and delays in processing customer requests. These limitations negatively affect both service providers and customers by reducing the speed and reliability of transactions.

In contrast, the implementation of web-based systems introduces automation and centralized database management, which significantly reduces manual workload and human errors while improving data organization and accessibility. Features such as online booking, real-time updates, and automated record management allow users and administrators to interact more efficiently within a unified system. Additionally, several studies emphasize that integrating structured development methodologies ensures better system design, improved functionality, and higher system reliability.

Overall, the findings of these local studies strongly support the idea that digital transformation is essential in modernizing service-based applications in the Philippines. The use of automated systems not only improves operational performance but also enhances user experience by providing faster, more convenient, and more reliable services. These insights serve as a strong foundation for the development of the proposed Car Rental with Cash Register System, as they highlight the importance of automation, centralized databases, and efficient transaction processing in achieving an effective and dependable system.

2.3 Related Literature

Web-Based Systems

Web-based systems have become one of the most significant technological innovations in modern information technology, providing organizations with efficient and accessible solutions for managing business operations and delivering services to users. A web-based system is a software application that operates through a web browser and can be accessed through a network or the internet without requiring installation on individual devices. These systems enable users to interact with applications and databases remotely, making them highly flexible, scalable, and convenient for both organizations and customers. As digital technology continues to advance, web-based systems have become widely adopted across various industries, including education, healthcare, banking, retail, tourism, transportation, and vehicle rental services.

The growing popularity of web-based systems can be attributed to their ability to provide real-time access to information and services. Unlike traditional desktop applications that are limited to specific computers or locations, web-based systems allow authorized users to access information from anywhere with an internet connection. This accessibility improves communication, coordination, and operational efficiency within organizations. Employees, administrators, and customers can perform transactions, retrieve information, and manage records through a centralized platform without being physically present in a particular location. As a result, web-based systems support more flexible and responsive business operations.

One of the primary advantages of web-based systems is their ability to automate business processes. Traditional manual systems often require employees to perform repetitive tasks such as recording transactions, updating records, calculating payments, and preparing reports. These processes consume significant amounts of time and increase the possibility of human error. Through automation, web-based systems can execute these tasks accurately and consistently while reducing administrative workload. Automated processing not only improves efficiency but also allows employees to focus on more important responsibilities that require critical thinking and customer interaction.

Another important feature of web-based systems is centralized data management. Information entered into the system is stored in a centralized database where it can be accessed, updated, and maintained by authorized users. Centralized databases help ensure data consistency and accuracy by eliminating duplicate records and reducing the risk of information loss. In service-oriented businesses, centralized data management enables efficient tracking of customer information, transactions, reservations, and operational records. This capability improves organizational productivity and supports better decision-making by providing accurate and updated information whenever needed.

Web-based systems also contribute significantly to improving customer satisfaction. Modern customers expect fast, convenient, and reliable services that can be accessed with minimal effort. Through web-based platforms, users can access services at any time and from any location, eliminating the need for unnecessary travel or lengthy waiting periods. Customers can view available services, make reservations, update information, and receive confirmations instantly. These features enhance convenience and create a more positive user experience, which is essential for maintaining customer trust and loyalty.

In addition to improving accessibility and convenience, web-based systems offer enhanced security and data protection. Modern systems incorporate various security measures such as user authentication, password encryption, role-based access control, and database backup mechanisms. These features help protect sensitive information from unauthorized access, data breaches, and accidental loss. Organizations that implement secure web-based systems are better equipped to safeguard customer information and maintain the integrity of their business operations.

Scalability is another major benefit associated with web-based systems. As organizations grow and operational requirements increase, web-based applications can be expanded and upgraded without requiring significant changes to existing infrastructure. New features, additional users, and larger volumes of data can be accommodated more easily compared to traditional systems. This flexibility makes web-based systems highly suitable for businesses that anticipate future growth and changing operational demands.

Furthermore, web-based systems support efficient reporting and monitoring capabilities. Information stored within the database can be analyzed and transformed into meaningful reports that assist managers and administrators in evaluating business performance. Automated report generation allows organizations to monitor transactions, identify trends, assess operational efficiency, and make informed decisions. The availability of accurate and timely reports contributes to improved planning, resource allocation, and strategic development.

In transportation and vehicle rental services, web-based systems play a crucial role in improving reservation management, vehicle monitoring, customer record handling, and transaction processing. Through a centralized and automated platform, administrators can monitor vehicle availability in real time, process reservations efficiently, and maintain accurate customer information. Customers also benefit from faster and more convenient booking procedures, resulting in improved service quality and overall satisfaction.

The relevance of web-based systems to the proposed **Cash Register and Reservation System for Calma Transportation Services** is highly significant. The proposed system utilizes web-based technology to automate reservation handling, payment processing, customer record management, and report generation. By replacing manual processes with a computerized platform, the system aims to improve operational efficiency, reduce human errors, enhance data security, and provide better service to customers. Consequently, the implementation of a web-based system serves as a practical and effective solution for addressing the operational challenges currently experienced by Calma Transportation Services while supporting its long-term growth and development.

System Automation

System automation refers to the use of technology to perform tasks and processes with minimal human intervention. In modern organizations, automation has become a critical component in improving operational efficiency, reducing repetitive workloads, and enhancing service delivery. Through automated systems, businesses can execute routine activities such as data entry, record management, transaction processing, and report generation more accurately and consistently than manual methods. As technology continues to evolve, automation has transformed the way organizations operate by enabling faster, more reliable, and more efficient business processes.

The implementation of automation provides numerous benefits to service-oriented industries. Traditional manual operations often require employees to spend significant time performing repetitive tasks, which can increase the likelihood of human errors and reduce overall productivity. Automated systems eliminate many of these challenges by executing predefined procedures consistently and accurately. This allows employees to focus on more important responsibilities that require decision-making and customer interaction. Consequently, automation contributes to improved efficiency, better resource utilization, and enhanced organizational performance.

In transportation and vehicle rental businesses, automation plays an important role in managing reservations, customer records, vehicle availability, and payment transactions. Automated reservation systems can instantly update booking information, preventing scheduling conflicts and reducing administrative workload. Similarly, automated payment processing ensures accurate calculations and faster transaction completion. These capabilities contribute to improved customer satisfaction and more efficient service delivery.

The proposed Cash Register and Reservation System incorporates automation to streamline reservation management, transaction processing, and record keeping. By reducing dependence on manual procedures, the system aims to improve operational efficiency, minimize errors, and provide a more reliable service experience for both administrators and customers.

Database Management

Database management is a fundamental component of modern information systems that involves the organization, storage, retrieval, and maintenance of data within a structured environment. As businesses generate increasing amounts of information, effective database management has become essential in ensuring that data remains accurate, accessible, and secure. Databases serve as centralized repositories where information can be stored systematically and accessed whenever needed by authorized users.

Traditional methods of storing information often rely on paper records, spreadsheets, and separate files. While these methods may be suitable for small-scale operations, they become inefficient as data volume increases. Manual record management can lead to data duplication, inconsistencies, loss of information, and difficulties in retrieving records. Database management systems address these challenges by providing structured storage mechanisms that improve data integrity and accessibility.

Centralized database systems allow organizations to maintain a single source of information that can be shared across multiple functions and departments. This ensures consistency and accuracy while reducing redundancy. In reservation-based businesses, databases play a crucial role in maintaining customer information, booking records, vehicle details, and payment histories. Real-time updates ensure that information remains current and available whenever needed.

The proposed system utilizes a centralized database to manage reservations, customer profiles, vehicles, and transactions efficiently. Through proper database management, the system ensures accurate information processing, improved accessibility, and enhanced operational performance.

User Interface Design

User Interface (UI) Design refers to the process of creating visual and interactive elements that enable users to communicate effectively with a system. A well-designed interface serves as the bridge between users and technology by presenting information in a clear, organized, and understandable manner. The success of any information system largely depends on how easily users can navigate and interact with its features.

Modern literature emphasizes that usability is one of the most important factors influencing user satisfaction and system acceptance. Systems that are difficult to understand or navigate often discourage users and reduce productivity. Effective user interface design focuses on simplicity, consistency, responsiveness, and accessibility. These principles help users complete tasks efficiently while minimizing confusion and errors.

In web-based systems, user interfaces must accommodate individuals with varying levels of technical expertise. Features such as intuitive navigation menus, clearly labeled functions, organized layouts, and responsive designs contribute to a positive user experience. A well-designed interface not only improves efficiency but also increases user confidence in using the system.

For the proposed Cash Register and Reservation System, user interface design is a critical consideration. The system aims to provide an accessible and user-friendly environment that enables administrators and staff members to manage reservations, payments, and records efficiently.

Cash Register Systems

Cash Register Systems are computerized solutions designed to record, calculate, and manage financial transactions within a business. These systems have evolved significantly from traditional mechanical cash registers into sophisticated digital platforms capable of handling sales, payments, inventory updates, and financial reporting. In modern business environments, cash register systems play an essential role in ensuring transaction accuracy and financial accountability.

Manual transaction processing often involves handwritten receipts and manual calculations, which can lead to errors and inconsistencies. These mistakes may result in inaccurate financial records, customer disputes, and operational inefficiencies. Cash register systems address these issues by automating payment calculations and maintaining detailed transaction histories.

Modern cash register systems are commonly integrated with databases and reporting tools, allowing businesses to monitor sales activities and generate financial summaries efficiently. This capability supports better financial management and informed decision-making. Additionally, automated systems reduce administrative workload and improve transaction speed.

The proposed system integrates a cash register feature that automates payment calculations and transaction recording. This functionality ensures accurate financial processing while supporting efficient business operations.

System Development Life Cycle (SDLC)

The System Development Life Cycle (SDLC) is a structured methodology used in the planning, design, development, testing, implementation, and maintenance of information systems. SDLC provides a systematic approach that helps developers create reliable, efficient, and user-centered systems while minimizing development risks and ensuring project success.

The SDLC process typically begins with planning and requirements gathering, where developers identify organizational needs and system objectives. This is followed by analysis and design phases, during which system specifications and architectures are developed. Once the design is completed, developers proceed to implementation, testing, deployment, and maintenance activities.

One of the major advantages of SDLC is its ability to ensure that systems are developed according to user requirements and organizational goals. Structured development processes improve quality assurance and reduce the likelihood of system failures. Additionally, SDLC provides a framework for evaluating project progress and maintaining system performance after deployment.

The development of the proposed Cash Register and Reservation System follows SDLC principles to ensure that the final product meets functional requirements and supports efficient business operations.

Reservation Management System

Reservation Management Systems are computerized platforms designed to facilitate the scheduling, monitoring, and management of reservations in service-oriented businesses. These systems allow organizations to efficiently handle customer bookings while minimizing manual intervention. Traditionally, reservation processes relied on paper records, phone calls, and face-to-face transactions, which often resulted in scheduling conflicts, delayed confirmations, and data inaccuracies. With the advancement of information technology, reservation systems have evolved into web-based platforms capable of providing real-time updates and automated transaction processing.

Modern reservation systems provide numerous benefits to both service providers and customers. They improve operational efficiency by automating booking procedures, reducing paperwork, and maintaining accurate records of reservations. Customers benefit from faster access to services and the ability to make reservations at their convenience. In transportation-related industries, reservation systems play a significant role in ensuring that vehicle availability is properly monitored and that rental schedules are managed effectively.

Furthermore, reservation systems contribute to improved customer satisfaction by reducing waiting times and providing more reliable services. Through centralized databases, businesses can easily retrieve customer information, track reservation history, and generate reports that support decision-making. As a result, reservation management systems have become an essential component of modern service operations.

Management Information System

Management Information Systems (MIS) refer to organized systems that collect, process, store, and distribute information to support decision-making, coordination, and control within an organization. MIS serves as a bridge between technology and business operations by providing managers and employees with accurate and timely information necessary for efficient organizational performance.

In service-oriented businesses, MIS improves operational effectiveness by integrating various business functions into a centralized platform. These systems enable organizations to manage customer records, monitor transactions, generate reports, and analyze business performance more effectively. Through the use of computerized systems, businesses can reduce the time required to process information and improve the accuracy of operational activities.

The implementation of MIS also contributes to better communication and coordination among employees. Since information is stored in a centralized database, authorized users can easily access updated records whenever necessary. This reduces duplication of work, minimizes data inconsistencies, and ensures that business decisions are based on accurate information.

Furthermore, MIS supports strategic planning by providing managers with relevant reports and performance indicators. These reports help organizations identify trends, evaluate operational efficiency, and develop appropriate strategies for future growth. In transportation-related businesses, management information systems play a vital role in monitoring reservations, vehicle utilization, customer transactions, and financial records.

As businesses continue to embrace digital transformation, MIS remains an essential component for improving efficiency, enhancing productivity, and maintaining competitiveness in a rapidly changing business environment.

Customer Relationship Management

Customer Relationship Management (CRM) is a business strategy and technological approach that focuses on building and maintaining strong relationships between organizations and their customers. CRM systems are designed to collect, organize, and analyze customer information to improve communication, customer service, and overall customer satisfaction.

In service industries, customer satisfaction is one of the most important factors influencing business success. Customers expect fast, reliable, and convenient services that meet their needs efficiently. CRM systems help organizations achieve these expectations by providing detailed information about customer preferences, transaction histories, and service interactions.

The use of CRM technology allows businesses to personalize their services and improve customer engagement. By maintaining accurate customer records, organizations can better understand customer needs and provide services that match their expectations. This leads to increased customer loyalty and stronger business relationships.

CRM systems also improve communication by enabling businesses to respond to customer inquiries, complaints, and feedback more effectively. Through organized customer databases, service providers can quickly access relevant information and resolve issues in a timely manner.

In the context of transportation and car rental services, CRM plays a significant role in managing customer reservations, monitoring rental history, and maintaining customer records. The integration of CRM principles into a reservation system can improve customer experience by providing more personalized, efficient, and reliable services.

Digital Transformation in Transportation Services

Digital transformation refers to the integration of digital technologies into various business processes to improve efficiency, productivity, and service quality. In recent years, transportation services have undergone significant digital transformation due to advances in information technology, internet connectivity, and mobile applications.

Traditional transportation businesses often relied on manual procedures for managing reservations, scheduling services, and recording transactions. These processes were frequently associated with delays, inefficiencies, and human errors. Through digital transformation, transportation companies have adopted computerized systems that automate many of these operational activities.

One of the major benefits of digital transformation is the ability to provide real-time information. Customers can easily access transportation services, check availability, make reservations, and receive updates through digital platforms. This improves convenience and enhances the overall customer experience.

For transportation service providers, digital technologies improve operational management by enabling real-time monitoring of resources, automated transaction processing, and centralized data management. These capabilities contribute to more efficient service delivery and better resource utilization.

Moreover, digital transformation supports business growth by allowing organizations to expand their services and reach a wider customer base. Web-based platforms and online reservation systems provide businesses with opportunities to improve accessibility while reducing operational costs.

As transportation services continue to evolve, digital transformation remains a critical factor in improving competitiveness, efficiency, and customer satisfaction within the industry.

Data Security and Privacy

Data security and privacy are important considerations in the development and implementation of information systems. As organizations increasingly rely on digital platforms to manage sensitive information, protecting data from unauthorized access, loss, and misuse has become a major priority.

Data security refers to the measures and technologies used to safeguard information against threats such as hacking, data breaches, malware attacks, and unauthorized modifications. Effective security practices ensure that information remains confidential, accurate, and accessible only to authorized users.

Privacy, on the other hand, focuses on protecting personal information and ensuring that customer data is collected, stored, and used responsibly. Organizations have a responsibility to maintain the confidentiality of customer information and comply with applicable data protection regulations.

In web-based systems, several security measures are commonly implemented to protect data. These include user authentication, password encryption, access control mechanisms, database security protocols, and regular system backups. Such measures reduce the risk of data loss and unauthorized access.

For transportation and car rental services, protecting customer information is particularly important because systems often store personal details, booking records, and payment information. Failure to secure this data can lead to financial losses, reputational damage, and reduced customer trust.

Therefore, incorporating data security and privacy measures into system design is essential for ensuring system reliability, protecting user information, and maintaining customer confidence. The proposed Cash Register and Reservation System recognizes the importance of these principles and aims to provide a secure environment for managing customer and transaction data.

2.4 Synthesis of the Reviewed Literature

The reviewed literature and studies, both foreign and local, collectively emphasize the growing importance and relevance of web-based systems in improving operational efficiency, accuracy, reliability, and overall service delivery across various industries, particularly in transportation and vehicle rental services. Across all examined studies, a consistent and recurring theme emerges: traditional manual processes are no longer sufficient to meet the increasing demands of modern service-oriented environments due to their inherent limitations in speed, accuracy, accessibility, and data management capabilities. As business operations become more complex and customer expectations continue to rise, the need for automated and integrated digital solutions becomes increasingly essential.

Foreign studies strongly highlight that car rental systems and similar transportation platforms benefit significantly from the integration of automation, real-time communication, and advanced system functionalities such as GPS tracking, intelligent scheduling algorithms, mobile application integration, and dynamic booking systems. These technologies enhance operational workflows by reducing human intervention, minimizing errors, and ensuring that data is processed, updated, and synchronized in real time. Furthermore, foreign literature emphasizes that the integration of multiple system components—such as booking, payment processing, monitoring, and reporting—into a unified platform leads to improved decision-making, better resource management, and enhanced operational control. This ultimately results in higher levels of customer satisfaction and more efficient business operations.

On the other hand, local studies conducted in the Philippines reinforce and support these findings by demonstrating that many service-based businesses, including car rental and transportation services, still rely heavily on manual or semi-digital systems. These traditional approaches often result in operational inefficiencies such as booking conflicts, delayed transaction processing, inconsistent record keeping, and difficulty in tracking real-time vehicle availability. The reviewed local literature consistently shows that the implementation of web-based systems significantly improves transaction processing speed, data accuracy, and service accessibility. In addition, automation and centralized database systems are repeatedly identified as critical factors in reducing administrative workload, minimizing human error, and enhancing overall operational performance within local business environments.

Moreover, both foreign and local studies highlight the importance of system usability, accessibility, and structured development methodologies in ensuring system effectiveness and sustainability. Features such as user-friendly interfaces, responsive design, real-time updates, secure data handling, and integrated financial transaction systems are consistently recognized as essential components of an effective web-based application. The inclusion of cash register or payment management functionalities is also emphasized in related literature as a vital feature for ensuring accurate financial tracking, transparent transaction processing, and improved accountability in business operations. Additionally, studies stress the importance of following structured development approaches such as Agile or SDLC to ensure systematic planning, continuous improvement, and successful system implementation.

Furthermore, system security and data integrity are also highlighted as important considerations in modern literature. As web-based systems handle sensitive information such as customer data, booking details, and financial transactions, implementing proper security measures such as authentication, authorization, and encrypted data storage is essential to protect system integrity and user privacy. Scalability and system performance are also important aspects, especially in systems that must accommodate increasing numbers of users and transactions over time without compromising speed or efficiency.

Overall, the synthesis of the reviewed literature and studies clearly indicates that the integration of automation, centralized database systems, real-time processing, secure data management, and financial transaction handling is essential in developing efficient, reliable, and user-centered service-based applications. These combined insights strongly support the development of the proposed **Car Rental with Cash Register and Reservation System**, as it aims to address the limitations of traditional manual processes by providing a unified, automated, secure, and user-friendly platform that enhances both operational efficiency and customer experience.

CHAPTER 3: METHODOLOGY

3.1 Research Design

This study employed a **Developmental Research Design** as the primary research approach in developing the proposed *Cash Register and Reservation System for Calma Transportation Services*. This type of research design is commonly used in information technology and software development projects because it focuses on creating, improving, and evaluating a product, system, or application that addresses existing operational problems. Unlike other research designs that mainly concentrate on describing or analyzing a situation, developmental research emphasizes the development of practical solutions that can be implemented in real-world environments.

The researchers selected this research design because the main objective of the study is not only to examine the current reservation and transaction management practices of Calma Transportation Services but also to develop a computerized system that can improve these processes. At present, many businesses still encounter challenges related to manual record keeping, reservation monitoring, customer management, and transaction processing. These challenges often result in delays, inaccuracies, data redundancy, and difficulties in generating reports. Through developmental research, the researchers were able to systematically identify these concerns and develop a technology-based solution that addresses the specific needs of the business.

The developmental research process began with a thorough assessment of the existing operational procedures used by Calma Transportation Services. The researchers conducted interviews and observations to gain a clear understanding of how reservations are processed, how customer information is recorded, how transactions are handled, and how business records are maintained. This initial investigation enabled the researchers to identify weaknesses and inefficiencies within the current workflow and determine areas that required improvement through automation.

After gathering and analyzing the necessary information, the researchers proceeded with the design and development of the proposed system. The development process involved identifying system requirements, designing system components, creating the database structure, developing user interfaces, implementing system functionalities, and conducting testing procedures. Each stage was carefully planned and executed to ensure that the final product would effectively support the operational requirements of the business and provide a more efficient method of managing reservations and transactions.

Another important characteristic of developmental research is its emphasis on continuous improvement throughout the development process. As the system was being developed, feedback and observations were continuously gathered to evaluate whether the system met its intended objectives. Necessary modifications and enhancements were implemented whenever issues or limitations were identified. This iterative process allowed the researchers to improve the overall quality, functionality, and usability of the system before its final evaluation.

Furthermore, developmental research encourages the integration of theoretical knowledge and practical application. In this study, concepts and principles related to information systems, database management, software engineering, and web application development were applied to create a system that addresses actual business challenges. By combining academic knowledge with practical implementation, the researchers were able to develop a solution that is both technically sound and operationally effective.

The use of developmental research design also supports the evaluation of the completed system. After development, the proposed Cash Register and Reservation System underwent testing and assessment to determine its effectiveness, reliability, usability, and functionality. The results of these evaluations provided valuable insights into the strengths of the system and identified potential areas for future improvement. This ensured that the developed application met the expectations of its intended users and fulfilled the objectives established at the beginning of the study.

Through the application of Developmental Research Design, the researchers were able to follow a structured and systematic process from problem identification to solution development and evaluation. This approach contributed significantly to the successful creation of a web-based Cash Register and Reservation System capable of improving reservation management, transaction processing, record keeping, and overall operational efficiency within Calma Transportation Services. Ultimately, the study demonstrates how technology can be utilized as a practical tool for enhancing business operations and delivering better service to customers.

3.2 System Development Methodology

The researchers utilized the **System Development Life Cycle (SDLC)** as the primary methodology for the development of the proposed *Cash Register and Reservation System for Calma Transportation Services*. SDLC is a widely recognized framework in software development that provides a systematic and organized approach to creating information systems. It consists of several phases that guide developers from the initial identification of a problem up to the deployment and maintenance of the finished product. By following this methodology, the researchers were able to ensure that every aspect of the system was carefully planned, analyzed, developed, tested, and evaluated before implementation.

The selection of SDLC was based on its ability to provide a clear roadmap throughout the development process. Since the proposed system involves managing reservations, customer records, vehicle availability, and financial transactions, a structured methodology was necessary to ensure that all requirements were properly addressed. SDLC also helps reduce development risks by allowing developers to identify problems early in the process and make necessary adjustments before the system reaches its final stage. Through the use of this methodology, the researchers were able to develop a system that is reliable, efficient, and capable of meeting the operational needs of Calma Transportation Services.

Planning Phase

The planning phase served as the foundation of the entire development process. During this stage, the researchers conducted preliminary investigations to gain a thorough understanding of the current operational procedures of Calma Transportation Services. The objective was to identify existing challenges and determine how technology could be utilized to improve business operations. Interviews and observations were conducted to gather information regarding reservation handling, customer management, vehicle scheduling, transaction processing, and record keeping.

The researchers carefully examined the limitations of the current system and identified several areas that required improvement. Common issues included delays in processing reservations, difficulties in monitoring vehicle availability, manual recording of transactions, and challenges in maintaining accurate business records. These findings helped establish the objectives of the study and provided direction for the development of the proposed system.

In addition, project requirements, timelines, development resources, and expected outcomes were determined during this phase. Proper planning allowed the researchers to establish realistic goals and ensure that the project would remain organized throughout the development process.

Analysis Phase

The analysis phase focused on examining the information gathered during the planning stage in greater detail. The researchers analyzed the existing workflow and identified the specific requirements needed to address the problems experienced by the business. This phase involved studying how information flows within the organization and determining how each process could be improved through automation.

Functional requirements were identified to specify the tasks that the system must be capable of performing. These requirements included customer registration, reservation processing, vehicle management, payment recording, report generation, and user account administration. The researchers also identified non-functional requirements such as system security, reliability, performance, accessibility, and ease of use.

Through careful analysis, the researchers were able to define the scope and limitations of the system. This ensured that the project remained focused on developing features that directly contribute to improving reservation management and cash register operations. The results of this phase served as the basis for designing the overall structure and functionality of the proposed application.

Design Phase

Once all requirements were clearly identified, the researchers proceeded to the design phase. This stage focused on creating a detailed blueprint that would guide the development of the system. The design process involved determining how the system would operate, how information would be stored, and how users would interact with the application.

Several design tools and techniques were utilized during this phase, including Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), Use Case Diagrams, database schemas, and interface prototypes. These diagrams provided visual representations of the system's structure and functionality, making it easier to understand how different components would interact with one another.

The researchers also focused on designing an intuitive and user-friendly interface. Since both administrators and customers would interact with the system, special consideration was given to ensuring that menus, forms, buttons, and navigation features were easy to understand and use. The design phase played a crucial role in ensuring that the final application would be both functional and visually appealing.

Development Phase

The development phase involved transforming the approved system design into a fully functional web-based application. During this stage, the researchers utilized various programming languages, frameworks, and development tools to implement the planned features and functionalities of the system.

The frontend interface was developed to provide users with an interactive environment where they could make reservations, view booking information, manage accounts, and access available services. At the same time, the backend system was developed to handle business logic, data processing, authentication, and communication with the database.

Several core features were implemented during this phase, including reservation management, customer account management, vehicle monitoring, payment recording, report generation, and administrative controls. The researchers ensured that each component was developed according to the specifications established during the design phase. Continuous monitoring and debugging were also performed to minimize coding errors and improve system performance.

This phase represented the most time-consuming part of the project because it required translating theoretical designs into actual software components. However, it was also one of the most important stages because it transformed the researchers' ideas into a practical solution capable of addressing real business challenges.

Testing Phase

After the system was developed, extensive testing procedures were conducted to ensure that all functionalities worked as intended. The primary purpose of testing was to identify errors, defects, and inconsistencies that could affect the performance of the system.

The researchers performed different testing activities to verify the accuracy of system outputs and ensure that all modules functioned properly. Reservation processing, transaction recording, report generation, user authentication, and database operations were carefully examined. The testing process also included evaluating how the system responded to various user inputs and operational scenarios.

Whenever errors or issues were discovered, corrective actions were immediately taken. The affected components were revised and tested again until the desired results were achieved. Through repeated testing and validation, the researchers ensured that the final system was stable, reliable, and capable of performing its intended functions effectively.

Implementation Phase

Following successful testing, the system proceeded to the implementation phase. During this stage, the completed application was deployed and introduced to its intended users for demonstration and evaluation purposes. Administrators and staff members were given the opportunity to interact with the system and perform actual tasks using the developed application.

The implementation phase allowed users to experience the features of the system in a realistic environment. Users evaluated the system's performance, functionality, ease of use, and overall effectiveness in supporting business operations. Their feedback provided valuable insights regarding the strengths and weaknesses of the application.

This phase was important because it determined whether the developed system could successfully meet user expectations and address the operational problems identified during the earlier stages of the project.

Maintenance Phase

The maintenance phase is the final stage of the System Development Life Cycle. Although the proposed system was developed primarily for academic purposes, maintenance remains an important consideration for future use and enhancement. Technology and business requirements continuously evolve, making it necessary for software applications to be updated and improved over time.

Maintenance activities may include correcting software bugs, improving security features, optimizing system performance, and adding new functionalities based on user feedback. As Calma Transportation Services expands its operations, additional features such as online payment integration, automated notifications, GPS tracking, and advanced analytics may be incorporated into future versions of the system.

Regular maintenance helps ensure that the system remains reliable, secure, and effective in supporting daily business operations. It also extends the lifespan of the application and enables it to adapt to changing organizational needs and technological advancements.

Through the use of the System Development Life Cycle methodology, the researchers were able to follow a structured and systematic process in developing the proposed Cash Register and Reservation System. Each phase contributed significantly to the successful completion of the project, resulting in a web-based solution designed to improve efficiency, accuracy, and overall service quality within Calma Transportation Services.

3.3 System Architecture

The proposed *Cash Register and Reservation System for Calma Transportation Services* will utilize a **client-server architecture**, a widely used system structure in modern web applications. This architecture was selected because it provides an efficient way of managing communication between users, the application itself, and the database where important records are stored. Through this approach, data can be processed accurately, stored securely, and retrieved efficiently whenever needed. The architecture also supports multiple users accessing the system simultaneously without significantly affecting overall performance.

The client-server architecture consists of three major components: the **client side**, the **server side**, and the **database layer**. Each component performs a specific function that contributes to the overall operation of the system. By separating responsibilities among these components, the system becomes easier to manage, maintain, and improve in the future.

Client Side

The client side serves as the part of the system that users directly interact with. It provides the graphical user interface through which customers, administrators, and staff members can access the system's various features and services. Since the proposed system is web-based, users can access it using devices such as desktop computers, laptops, tablets, or smartphones with an internet connection and a compatible web browser.

For customers, the client side allows them to browse available transportation services and vehicles, submit reservation requests, view reservation details, and monitor the status of their bookings. The interface is designed to be simple, organized, and easy to navigate, ensuring that users can perform tasks efficiently even with minimal technical knowledge.

For administrators and staff members, the client interface provides access to management functions such as monitoring reservations, updating vehicle information, recording transactions, managing customer accounts, and generating reports. Through these features, authorized personnel can efficiently oversee daily operations and ensure that business processes run smoothly.

To improve accessibility and user experience, the interface is designed to be responsive. This means that the layout and functionality of the system automatically adjust according to the screen size and device being used. As a result, users can conveniently access the system from various devices without experiencing usability issues.

Server Side

The server side functions as the central processing unit of the system. It is responsible for handling requests received from users, processing information according to established business rules, and communicating with the database. Whenever a user performs an action, such as creating a reservation or recording a payment, the request is sent to the server for processing.

The server validates all incoming data to ensure accuracy and consistency before any information is stored in the database. For example, when a customer submits a reservation request, the server checks vehicle availability, verifies booking schedules, calculates applicable charges, and confirms whether the reservation can proceed. This process helps prevent scheduling conflicts and ensures that only valid transactions are recorded.

In addition, the server manages user authentication and access control. It verifies login credentials and determines the level of access granted to each user. This security mechanism ensures that sensitive functions, such as modifying reservations or generating financial reports, can only be performed by authorized personnel.

The server also handles communication between the user interface and the database. By acting as an intermediary, it helps protect sensitive information and reduces the risk of unauthorized access or data manipulation.

Database Layer

The database layer serves as the storage component of the system. It contains all information required for the daily operations of Calma Transportation Services. This includes customer records, reservation details, vehicle information, transaction histories, payment records, user accounts, and system-generated reports.

The database is designed using a relational structure to ensure that data is organized logically and efficiently. Related information is stored in separate tables and connected through relationships, reducing duplication and improving data consistency. This structure allows the system to retrieve information quickly while maintaining accuracy and integrity.

Whenever users create reservations, update records, or perform transactions, the information is automatically stored in the database. Similarly, when reports or booking details are requested, the database provides the required information to the server for processing and display.

The database also plays an important role in maintaining business records. Since information is stored electronically, administrators can easily search, retrieve, and generate reports without manually reviewing paper-based documents. This significantly improves productivity and reduces the likelihood of errors associated with manual record keeping.

System Communication Process

The interaction between the client side, server side, and database follows a continuous communication cycle. When a user submits a request through the client interface, the request is transmitted to the server. The server then processes the request and communicates with the database whenever information needs to be stored, modified, or retrieved.

After the necessary operations have been completed, the server sends the processed results back to the client side, where the information is displayed to the user. This process occurs within seconds, allowing users to receive immediate feedback regarding their actions.

For example, when a customer creates a reservation, the system follows the following sequence:

1. The customer enters reservation details through the web interface.
2. The information is sent to the server for validation.
3. The server checks vehicle availability and booking schedules.
4. The server records the reservation in the database.

5. The database confirms successful storage of the information.
6. The server sends a confirmation message to the user.
7. The reservation details are displayed on the customer's screen.

This workflow ensures that all transactions are processed accurately and efficiently while maintaining data consistency throughout the system.

Security and Scalability

One of the major advantages of using a client-server architecture is enhanced security. Sensitive operations such as user authentication, reservation validation, and transaction processing are performed on the server side rather than on the user's device. This reduces the possibility of unauthorized modifications and helps protect important business information.

The architecture also supports scalability, allowing the system to accommodate future growth. As the business expands and the number of users increases, additional resources can be added to the server and database without requiring major changes to the overall structure of the application. This flexibility ensures that the system remains capable of supporting future operational demands.

Overall, the client-server architecture provides a strong and reliable foundation for the proposed Cash Register and Reservation System. By separating the responsibilities of the client interface, server processing, and database management, the system can operate efficiently while maintaining security, reliability, and data accuracy. This architectural design supports the primary objective of improving reservation management, transaction processing, and record keeping within Calma Transportation Services while providing a convenient and user-friendly experience for both customers and administrators.

3.4 Tools and Technologies Used

The successful development of the proposed *Cash Register and Reservation System for Calma Transportation Services* requires the use of various software tools, programming technologies, and development environments. These technologies were carefully selected based on their capability to support the requirements of the system, including functionality, performance, security, scalability, and ease of maintenance. By utilizing modern web development technologies, the researchers were able to create a system that is efficient, user-friendly, and capable of supporting the operational needs of the business.

The development process involves both frontend and backend technologies, along with database management systems and development tools. Each technology plays a specific role in ensuring that the system operates effectively and delivers a smooth user experience.

Frontend

The frontend refers to the part of the system that users directly interact with through their web browsers. It is responsible for presenting information, collecting user inputs, and providing an interactive and visually appealing experience.

React with TypeScript

The researchers utilized **React** as the primary JavaScript library for developing the user interface of the system. React is widely recognized for its ability to create dynamic and responsive web applications through the use of reusable components. This approach allows developers to organize the application into smaller and manageable sections, making development more efficient and improving maintainability.

React was selected because it enables faster rendering of web pages and provides a seamless user experience. Since the proposed system involves frequent interactions such as viewing reservations, managing vehicle information, and processing transactions, React helps ensure that updates are displayed efficiently without requiring the entire page to reload.

Alongside React, **TypeScript** was used to improve code quality and reliability. TypeScript is an extension of JavaScript that introduces static typing, which helps developers identify potential errors during development rather than during system execution. This feature contributes to the stability of the application by reducing programming mistakes and improving overall code organization.

The combination of React and TypeScript provides a robust development environment that supports the creation of a modern, efficient, and scalable web application.

Vite

To support frontend development, the researchers utilized **Vite** as the build and development tool for the project. Vite is designed to provide faster startup times and quicker updates during development compared to traditional web development tools.

One of the primary advantages of Vite is its ability to improve developer productivity. It allows developers to immediately view changes made to the application without lengthy compilation processes. This feature significantly reduces development time and enables the researchers to test and refine system functionalities more efficiently.

Additionally, Vite produces optimized production builds, helping improve the overall performance and loading speed of the final application. Faster loading times contribute to a better user experience, especially for customers accessing the reservation system through different devices and internet connections.

Bootstrap

The researchers also utilized **Bootstrap** as the primary front-end framework for designing the visual appearance of the system. Bootstrap provides a collection of pre-designed user interface components such as buttons, forms, navigation menus, tables, and responsive layouts.

Bootstrap was selected because it simplifies the design process while maintaining consistency throughout the application. By using Bootstrap components, the researchers were able to create a professional-looking interface without developing every design element from scratch.

Another important advantage of Bootstrap is its responsive design capability. This ensures that the system automatically adjusts to different screen sizes, allowing users to access the application comfortably from desktop computers, laptops, tablets, and smartphones. Responsive design improves accessibility and ensures a positive user experience regardless of the device being used.

Backend

The backend serves as the operational core of the system. It processes user requests, executes business logic, validates information, and communicates with the database.

Node.js

For server-side development, the researchers utilized **Node.js** as the backend runtime environment. Node.js enables JavaScript code to run on the server, making it possible to handle system operations efficiently and manage communication between users and the database.

Node.js was selected because of its ability to process multiple requests simultaneously without significantly affecting system performance. This capability is particularly important for reservation systems where multiple users may attempt to access services at the same time.

Within the proposed system, Node.js is responsible for handling reservation requests, processing transactions, managing user authentication, validating information, and generating reports. It serves as the central component that coordinates interactions between the frontend and the database.

Furthermore, Node.js is known for its scalability and efficiency. As the number of users grows, the system can continue to perform effectively without major modifications to its architecture. This makes it a suitable technology for both current operations and future expansion.

Database

The database serves as the central repository for all information used within the system. Proper database management is essential for maintaining data accuracy, consistency, and security.

MySQL

The proposed system utilizes **MySQL** as its relational database management system. MySQL is one of the most widely used database platforms due to its reliability, efficiency, and ability to manage large amounts of structured data.

The database stores various types of information required for system operations, including customer records, reservation details, vehicle information, payment transactions, user accounts, and generated reports. By organizing data into related tables, MySQL helps reduce redundancy and ensures consistency across the entire system.

One of the major advantages of MySQL is its ability to retrieve information quickly and accurately. This capability is particularly important when users need to access reservation records, verify transaction histories, or generate business reports.

MySQL also provides security features that help protect sensitive information from unauthorized access. These features contribute to maintaining data integrity and ensuring that business records remain accurate and secure.

Development Environment

To facilitate efficient system development, the researchers utilized modern software development tools that support coding, debugging, testing, and project management activities.

Visual Studio Code (VS Code)

The primary Integrated Development Environment (IDE) used for the project was **Visual Studio Code (VS Code)**. VS Code is a lightweight yet powerful code editor widely used by professional developers and students alike.

VS Code provides numerous features that improve development efficiency, including syntax highlighting, code completion, debugging tools, extension support, and integrated version control. These features help developers write cleaner code, identify errors quickly, and maintain organized project files.

The software also supports various programming languages and frameworks used in the project, making it an ideal development environment for full-stack web application development. Through VS Code, the researchers were able to manage both frontend and backend components of the system within a single workspace.

3.5 Data Gathering Techniques

The success of the proposed *Cash Register and Reservation System for Calma Transportation Services* largely depends on the accuracy and relevance of the information gathered during the initial stages of the study. To ensure that the system addresses the actual needs and challenges faced by the business, the researchers employed appropriate data gathering techniques that provided valuable insights into the existing operational procedures. These techniques enabled the researchers to obtain reliable information regarding reservation management, customer transactions, vehicle monitoring, and record-keeping practices.

Data gathering is an important part of system development because it allows researchers to understand the current situation before proposing improvements. By collecting information directly from individuals involved in daily business operations, the researchers were able to identify existing problems, determine user requirements, and develop solutions that are practical and applicable to the organization.

For this study, the researchers utilized two primary data gathering techniques: **Interviews** and **Observation**. These methods were selected because they provide both qualitative and firsthand information regarding the actual processes and challenges experienced by Calma Transportation Services.

Interviews

Interviews were conducted to gather detailed information from individuals directly involved in the operations of Calma Transportation Services. This method allowed the researchers to communicate with business owners, employees, and other stakeholders who possess valuable knowledge regarding the organization's reservation procedures, transaction management, and customer service activities.

The interview process involved asking a series of prepared questions related to the current methods used in handling reservations, recording payments, managing customer information, and monitoring vehicle availability. In addition to structured questions, follow-up questions were also asked whenever clarification or additional details were needed. This approach enabled the researchers to gain a deeper understanding of the operational challenges encountered by the business.

Through the interviews, several important concerns were identified. These included difficulties in managing reservation records, delays in confirming bookings, challenges in monitoring vehicle availability, and the possibility of errors resulting from manual data entry. Stakeholders also expressed the need for a more organized and efficient system that could simplify daily operations and reduce the workload associated with manual processes.

The information gathered from interviews played a significant role in determining the features and functionalities that would be included in the proposed system. By listening to the experiences and suggestions of actual users, the researchers were able to develop a solution that directly addresses the needs of the organization.

Furthermore, interviews provided insights into user expectations regarding system usability and performance. Understanding these expectations helped ensure that the proposed application would not only function correctly but would also be easy to use and beneficial to its intended users. The interview process therefore served as a valuable source of information that guided both the design and development phases of the project.

Observation

In addition to interviews, the researchers utilized observation as a data gathering technique to obtain firsthand information regarding the actual workflow and business operations of Calma Transportation Services. Observation involves directly examining how tasks and activities are performed within the organization without relying solely on verbal descriptions from participants.

During the observation process, the researchers carefully monitored how reservations were received, recorded, and processed. They also observed how customer information was managed, how transactions were documented, and how vehicle availability was tracked. By witnessing these activities firsthand, the researchers gained a more accurate understanding of the operational environment and identified issues that may not have been fully revealed during interviews.

Observation allowed the researchers to identify inefficiencies within the existing process. For example, manual recording procedures often required additional time and effort, increasing the possibility of human errors and inconsistencies in data management. The researchers also observed situations where retrieving information from records became time-consuming due to the absence of a centralized system.

Another advantage of observation was the ability to verify information gathered during interviews. By comparing stakeholder responses with actual workplace practices, the researchers were able to ensure the accuracy and reliability of the collected data. This helped strengthen the overall validity of the study and provided a more comprehensive understanding of the organization's needs.

Additionally, observation enabled the researchers to identify opportunities for automation and process improvement. By understanding how employees performed their daily tasks, the researchers were able to determine which activities could benefit most from computerization. These observations contributed significantly to the design of system features intended to improve efficiency, reduce errors, and streamline business operations.

Importance of Combining Interviews and Observation

The researchers recognized that relying on a single data gathering technique may not provide a complete picture of the organization's operational requirements. For this reason, interviews and observation were used together to obtain more comprehensive and reliable information.

Interviews provided detailed explanations, opinions, and suggestions from stakeholders, while observation allowed the researchers to witness actual workplace practices and verify the accuracy of the information collected. The combination of these methods enabled the researchers to examine the problem from multiple perspectives and develop a more accurate understanding of the organization's needs.

Furthermore, using multiple data gathering techniques increased the credibility of the study. Information obtained through interviews could be validated through observation, reducing the possibility of misunderstandings or incomplete data. This approach ensured that the proposed system was developed based on factual information rather than assumptions.

The integration of these techniques also helped the researchers identify both technical and operational requirements for the system. As a result, the proposed Cash Register and Reservation System was designed to address real-world challenges while supporting the daily activities of Calma Transportation Services.

Overall, the use of interviews and observation provided the researchers with sufficient information to analyze existing processes, identify operational issues, and determine the most appropriate features for the proposed system. These data gathering techniques served as an essential foundation for the successful development of a web-based solution aimed at improving reservation management, transaction processing, record keeping, and overall service efficiency within Calma Transportation Services.

3.6 Testing Procedures

Testing is a crucial phase in system development because it ensures that the proposed system functions according to its intended purpose and meets the requirements identified during the earlier stages of development. It allows researchers to identify errors, inconsistencies, and technical issues before the system is fully implemented. Through proper testing, the reliability, functionality, and performance of the system can be evaluated and improved to provide a better experience for its intended users.

For the proposed *Cash Register and Reservation System for Calma Transportation Services*, testing was conducted to verify that all system components worked correctly and that the application was capable of supporting the operational needs of the business. The testing process focused on evaluating the system's ability to manage reservations, process transactions, maintain records, and provide accurate information to users.

The researchers conducted two major testing procedures, namely **System Testing** and **User Acceptance Testing (UAT)**. These testing methods were selected because they provide comprehensive evaluation of both the technical performance of the system and the satisfaction of its intended users.

System Testing

System Testing was conducted to evaluate the overall functionality of the developed application as a complete and integrated system. This testing procedure focused on determining whether all modules and components worked together properly and whether the system could perform its intended functions without errors or interruptions.

During this phase, the researchers tested various features and functionalities of the system, including user registration, login authentication, reservation management, vehicle information management, payment recording, report generation, and account administration. Each feature was carefully examined to ensure that it produced the expected results and handled user inputs correctly.

The researchers also tested how information moved between the frontend interface, backend processes, and database. This was done to verify that data was accurately transmitted, stored, retrieved, and displayed whenever users performed actions within the system. Proper communication among system components is essential to maintaining data consistency and ensuring reliable operation.

Several test scenarios were performed to evaluate the system under different conditions. These scenarios included valid and invalid user inputs, multiple reservation requests, transaction processing activities, and report generation procedures. By conducting these tests, the researchers were able to identify potential issues such as incorrect calculations, duplicate records, data retrieval errors, and interface inconsistencies.

Whenever problems were detected, corrective actions were implemented immediately. The affected components were modified and retested until they functioned correctly. This process helped ensure that the final version of the system was stable, accurate, and capable of supporting daily business operations effectively.

Another important objective of system testing was to evaluate the overall performance of the application. The researchers examined whether the system responded quickly to user requests and whether it could handle multiple operations without experiencing significant delays. Performance testing contributed to ensuring a smooth and efficient user experience.

Through comprehensive system testing, the researchers gained confidence that the proposed application met the required specifications and was ready for evaluation by actual users.

User Acceptance Testing (UAT)

Following the completion of system testing, the researchers conducted **User Acceptance Testing (UAT)** to determine whether the developed application satisfied the needs and expectations of its intended users. User Acceptance Testing is considered one of the most important stages of system evaluation because it involves actual users who will eventually utilize the system in real-world situations.

For this study, administrators, staff members, and selected individuals familiar with the operations of Calma Transportation Services participated in the evaluation process. Participants were given access to the system and were asked to perform various tasks such as creating reservations, managing bookings, recording transactions, viewing reports, and navigating different sections of the application.

The purpose of this testing phase was to assess whether users could successfully perform their required tasks using the system. It also allowed the researchers to evaluate the usability, functionality, and overall effectiveness of the application from the perspective of its end users.

During User Acceptance Testing, participants were encouraged to provide feedback regarding their experience with the system. They were asked to identify any difficulties encountered while using the application and to provide suggestions for improvement. Feedback was collected through evaluation forms, interviews, and informal discussions.

Particular attention was given to factors such as ease of use, interface design, accessibility of features, system responsiveness, and overall user satisfaction. The researchers analyzed the feedback received and used it to determine whether additional modifications or enhancements were necessary before finalizing the system.

User Acceptance Testing also helped validate whether the objectives established at the beginning of the study had been successfully achieved. If users were able to complete tasks efficiently and expressed satisfaction with the system's performance, it indicated that the application effectively addressed the operational challenges identified during the data gathering stage.

Importance of Testing in System Development

Testing plays a significant role in ensuring the quality and success of any software application. Without proper testing, errors and system failures may remain undetected, potentially affecting business operations and user satisfaction. By conducting thorough testing procedures, the researchers were able to identify weaknesses, improve system performance, and ensure that all functionalities operated as intended.

The testing process also contributes to the reliability and credibility of the study. Demonstrating that the system has undergone systematic evaluation provides assurance that it can perform effectively in actual operational environments. Furthermore, testing allows researchers to gather valuable feedback that can be used to improve future versions of the application.

For the proposed Cash Register and Reservation System, testing served as an essential step in validating the effectiveness of the developed solution. It ensured that the application was capable of improving reservation management, transaction processing, record keeping, and overall operational efficiency within Calma Transportation Services.

3.7 Evaluation Criteria

After the development and testing of the proposed *Cash Register and Reservation System for Calma Transportation Services*, the system will undergo evaluation to determine its effectiveness, quality, and suitability for actual business operations. The evaluation process is essential because it allows the researchers to measure whether the developed application successfully meets the objectives of the study and addresses the operational challenges identified during the data gathering phase.

The evaluation will focus on key software quality attributes that are commonly used in assessing information systems. These criteria help determine how well the system performs its intended functions, how easily users can interact with it, and how dependable it is during daily operations. Through these evaluation measures, the researchers can identify the strengths of the system as well as areas that may require further improvement.

The proposed system will be evaluated based on the following criteria: **Functionality, Usability, and Reliability**.

Functionality

Functionality refers to the ability of the system to perform its intended tasks accurately, efficiently, and completely. This criterion evaluates whether all required features and functions are working according to the specifications established during the planning and analysis stages of the project.

For the proposed Cash Register and Reservation System, functionality focuses on the system's ability to manage reservations, process transactions, maintain customer records, monitor vehicle availability, and generate reports. Each feature must operate correctly and provide accurate outputs based on user inputs.

The evaluation of functionality also determines whether the system effectively supports the daily workflow of Calma Transportation Services. For example, when a reservation is submitted, the system should accurately record the booking details, verify availability, and update the database without errors. Similarly, transaction records must be processed and stored correctly to ensure accurate financial reporting.

In addition, functionality evaluates the completeness of the system. This means that all required features identified during the requirement-gathering phase must be present and fully operational. Missing or malfunctioning features may negatively affect the efficiency and effectiveness of business operations.

A highly functional system enables users to perform their tasks more efficiently, minimizes operational errors, and contributes to improved service quality. Therefore, functionality serves as one of the most important indicators of the overall success of the proposed application.

Usability

Usability refers to the ease with which users can learn, understand, and operate the system. This criterion focuses on the overall user experience and evaluates whether the application is designed in a way that allows users to accomplish tasks efficiently and comfortably.

The proposed system is intended to be used by administrators, staff members, and customers with varying levels of technical knowledge. Because of this, the interface must be simple, organized, and user-friendly. Users should be able to navigate through different sections of the system without confusion or difficulty.

Several aspects are considered when evaluating usability. These include the clarity of menus and navigation options, the accessibility of features, the responsiveness of the interface, and the overall appearance of the application. Forms, buttons, and system messages should be easy to understand and guide users throughout the reservation and transaction processes.

Usability also measures how quickly users can become familiar with the system. A well-designed application should require minimal training and allow users to complete tasks efficiently even during their first interaction with the software. Reducing complexity and improving accessibility can significantly increase user satisfaction and productivity.

Furthermore, usability plays an important role in the acceptance of the system. Even if a system contains numerous features, users may hesitate to use it if it is difficult to understand or navigate. Therefore, ensuring a positive user experience is essential for the successful implementation of the proposed application.

Through usability evaluation, the researchers can determine whether the system provides a convenient and efficient platform for managing reservations, transactions, and business records.

Reliability

Reliability refers to the ability of the system to operate consistently and accurately over an extended period of time without experiencing failures, errors, or interruptions. This criterion evaluates the stability and dependability of the application during actual use.

For the proposed Cash Register and Reservation System, reliability is particularly important because business operations depend heavily on accurate records and uninterrupted access to information. Any system failure or data inconsistency could potentially affect reservation schedules, financial records, and customer satisfaction.

The evaluation of reliability focuses on the system's ability to maintain consistent performance under different operating conditions. The system should continue functioning properly even when handling multiple transactions, processing numerous reservations, or serving multiple users simultaneously.

Reliability also examines data accuracy and integrity. Information stored within the database must remain complete, consistent, and free from corruption. Reservation details, customer information, payment records, and generated reports should always reflect accurate and updated information.

Another important aspect of reliability is error handling. The system should be capable of identifying invalid inputs and responding appropriately without causing application crashes or loss of data. Effective error management contributes to a more stable and dependable user experience.

A reliable system increases user confidence and ensures that business operations can continue smoothly without unnecessary disruptions. For this reason, reliability serves as a critical factor in evaluating the overall quality of the developed application.

System Evaluation

System evaluation is an essential component of software development because it provides measurable evidence regarding the effectiveness of the proposed solution. Through evaluation, the researchers can determine whether the developed system successfully meets user requirements and fulfills the objectives established at the beginning of the study.

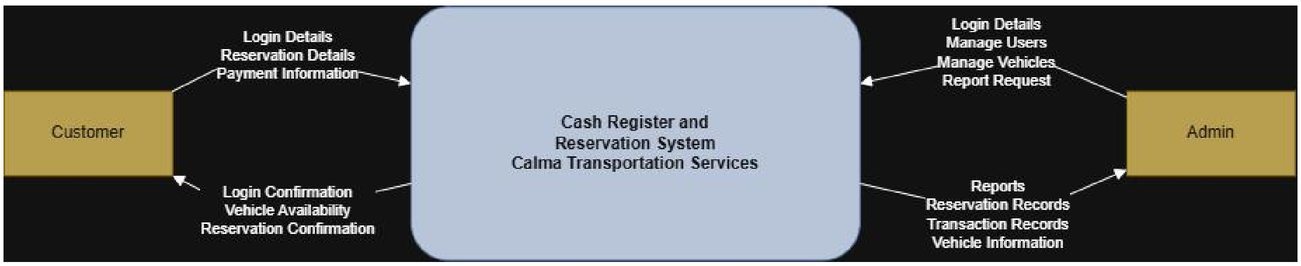
The results of the evaluation process may also provide valuable recommendations for future enhancements. Feedback obtained from users can help identify opportunities to improve system features, increase efficiency, strengthen security measures, and enhance overall performance.

Furthermore, evaluation ensures accountability within the development process by demonstrating that the system has been properly tested and assessed before implementation. This contributes to the credibility of the study and supports the validity of the research findings.

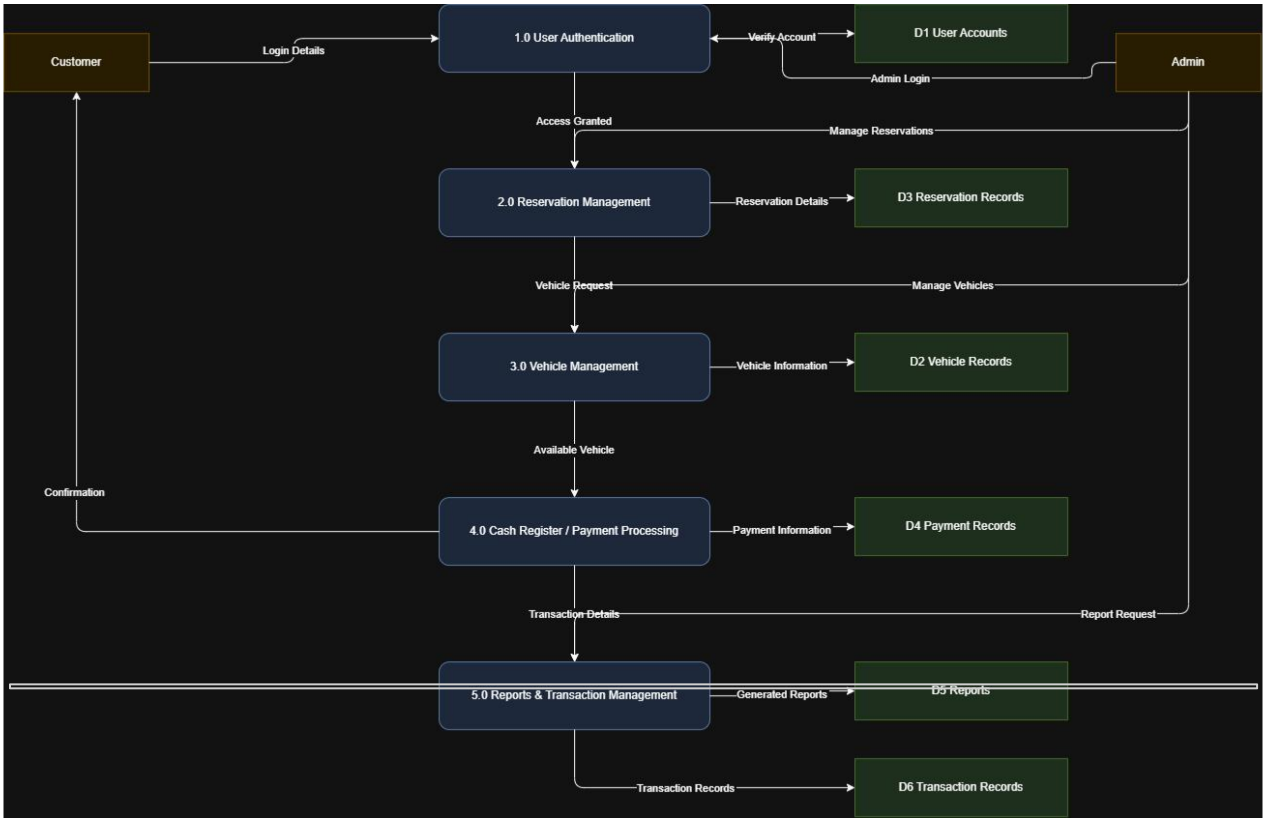
Overall, the evaluation of the proposed Cash Register and Reservation System based on functionality, usability, and reliability provides a comprehensive assessment of its quality and effectiveness. These criteria allow the researchers to measure the system's performance from both technical and user perspectives, ensuring that the developed application is capable of supporting the operational needs of Calma Transportation Services. Through this evaluation process, the researchers can determine whether the system successfully contributes to improving reservation management, transaction processing, record keeping, and overall business efficiency.

SUPPORTING DIAGRAMS

Data Flow Diagram (DFD – Level 0)



The **Data Flow Diagram (DFD) Level 0** presents a high-level overview of how information flows within the proposed Cash Register and Reservation System for Calma Transportation Services. It illustrates the interaction between the users, the system, and the database while showing how data is processed, stored, and retrieved during various operations.



The Data Flow Diagram (DFD) Level 1 provides a more detailed representation of the processes involved in the proposed Cash Register and Reservation System for Calma Transportation Services. It breaks down the main system into specific functions such as user authentication, reservation management, vehicle management, payment processing, cash register operations, and report generation. The diagram illustrates how data flows between users, the system, and the database during each process, showing how information is collected, processed, stored, updated, and retrieved to ensure efficient reservation handling, accurate payment recording, and reliable management of business operations.

Entity Relationship Diagram (ERD)

The **Entity Relationship Diagram (ERD)** illustrates the relationships between the various data entities used within the proposed Cash Register and Reservation System. It serves as the blueprint for the database structure and demonstrates how information is organized, connected, and managed throughout the application.

The database consists of three primary entities: **Users**, **Cars**, and **Reservations**.

Users Entity

The Users entity contains information about individuals who interact with the system. This includes administrators, staff members, and customers. Important attributes stored in this entity include:

- User ID
- Name
- Email Address
- Password
- Role

Cars Entity

The Cars entity contains information regarding the vehicles available for reservation. The system stores vehicle details to help customers select appropriate transportation options and enable administrators to manage available units effectively.

Attributes include:

- Car ID
- Car Name
- Brand
- Model
- Price Per Day
- Status
- Transmission Type
- Fuel Type
- Image URL

Reservations Entity

The Reservations entity stores information related to customer bookings. Each reservation record contains details necessary for monitoring and managing reservations.

Attributes include:

- Reservation ID
- Car Name
- Start Date
- End Date
- Total Price
- Status
- Created Date
- User ID

- Payment Method
- Reference Number

This entity serves as the link between customers and vehicles, allowing the system to track reservation history and transaction records.

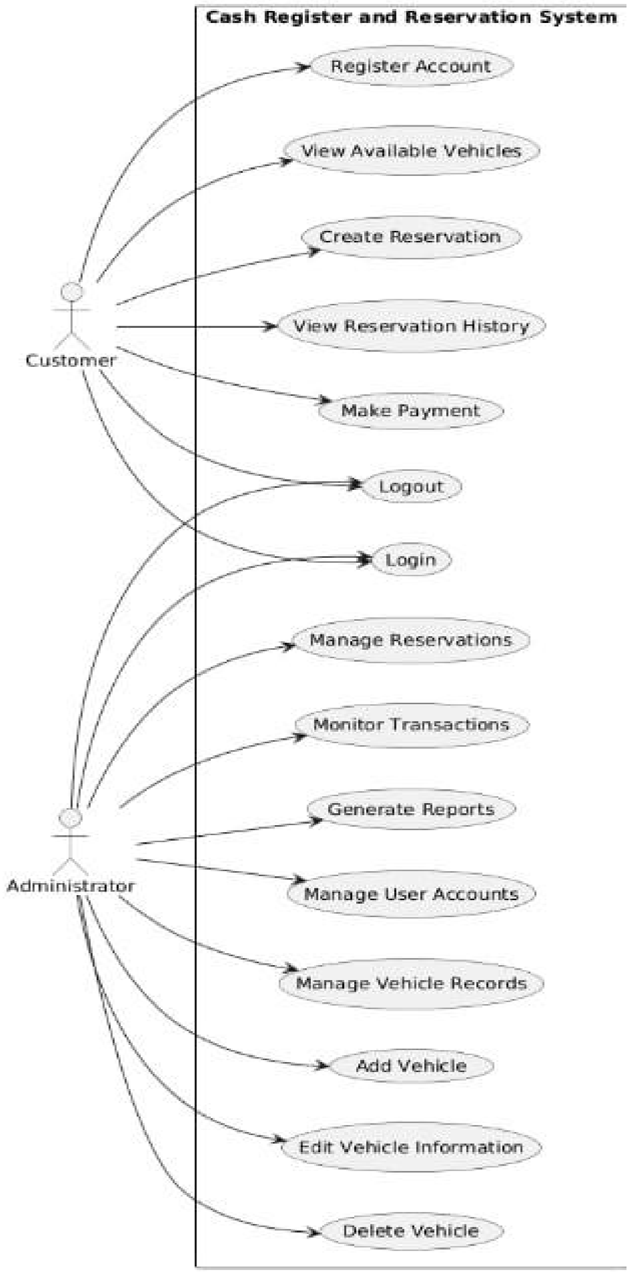
Relationships

The relationships between entities include:

- A User can create multiple Reservations.
- A Car can be associated with multiple Reservations over time.
- Administrators can manage Users, Cars, and Reservations.
- Each Reservation belongs to one User and one selected Car.

These relationships help maintain data consistency and ensure that reservation records are properly connected to both customers and vehicles.

Use Case Diagram



The **Use Case Diagram** illustrates the interactions between system users and the major functions available within the proposed Cash Register and Reservation System. It provides a visual representation of how users interact with the application and identifies the services offered by the system.

The diagram contains two primary actors:

User

The User represents customers who access the system to make reservations and manage their booking information.

The user can perform the following activities:

- Register an Account
- Login and Logout
- View Available Cars
- Create Reservations
- View Reservation History
- Make Payments through the Cash Register System

These functions allow customers to conveniently access transportation services and manage their reservations through a centralized platform.

Administrator

The Administrator is responsible for managing system operations and maintaining business records.

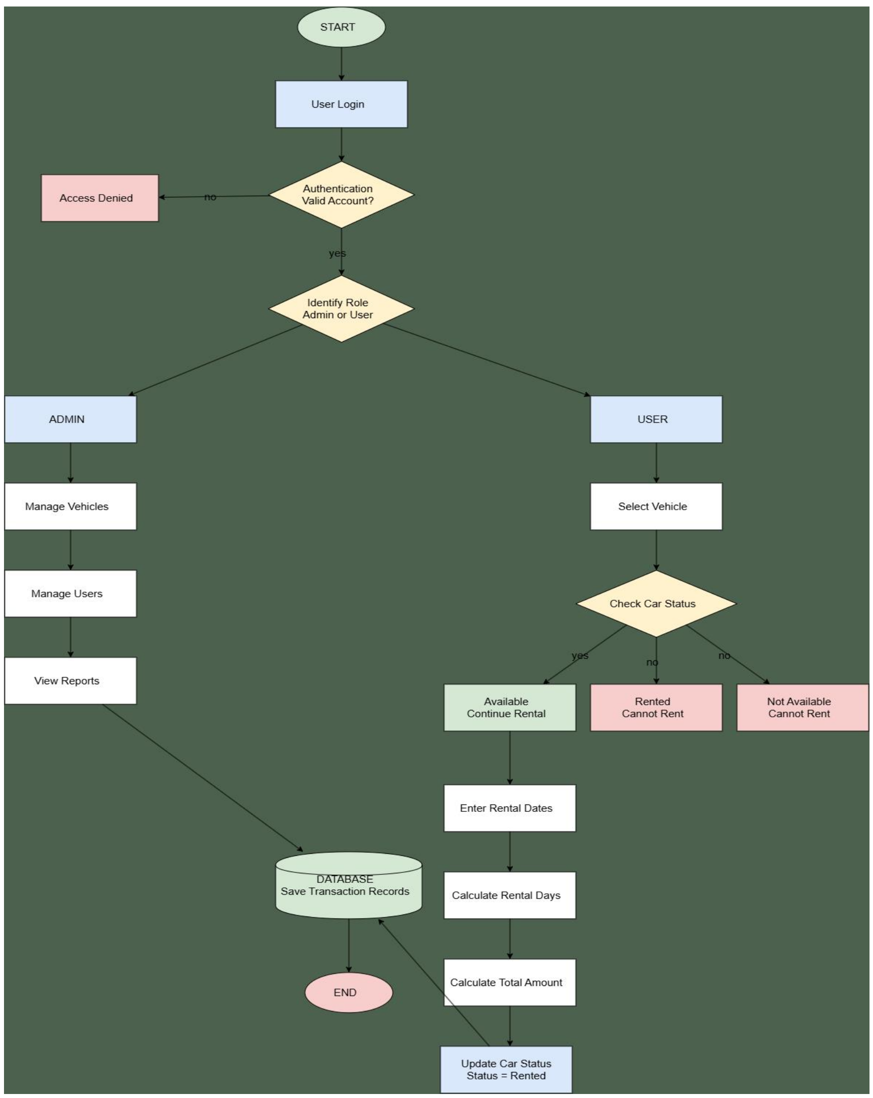
The administrator can perform the following activities:

- Login and Logout
- Manage Vehicle Records
- Add New Vehicles
- Edit Vehicle Information
- Delete Vehicle Records
- Manage Reservations
- Manage User Accounts
- Monitor Transactions
- Generate Reports

These administrative functions provide complete control over system operations and enable efficient management of transportation services.

CHAPTER 4 : SYSTEM DEVELOPMENT, IMPLEMENTATION TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 System Workflow Diagram



4.2 Overview of the Developed System

The **Calma Transportation Services Cash Register and Reservation System** was developed to improve the daily operations of Calma Transportation Services by replacing its existing manual reservation, payment, and record-keeping processes with an efficient computerized system. The developed system provides a centralized platform where administrators can manage customer reservations, vehicle information, payments, and reports accurately and in real time.

Prior to the development of the system, the business relied heavily on handwritten records and manual computation of rental charges. This traditional process often resulted in booking conflicts, inaccurate billing, misplaced records, and delays in serving customers. These issues affected the overall efficiency of the business and sometimes led to customer dissatisfaction. To address these challenges, the researchers developed a web-based application that automates the major business processes involved in transportation service management.

The system allows authorized administrators to securely access the application through a login page using registered credentials. Once authenticated, administrators can perform different tasks depending on their responsibilities. The dashboard provides an overview of important information such as the number of reservations, available vehicles, ongoing rentals, completed transactions, and system notifications. This enables administrators to quickly monitor the current status of business operations without manually checking multiple records.

One of the primary features of the developed system is the **reservation management module**. This module enables administrators to register customer information, create new reservations, update booking details, and monitor the availability of vehicles. The system automatically checks whether a selected vehicle is already reserved during a specific period, preventing double bookings and scheduling conflicts. This feature improves the accuracy of reservation management while minimizing human errors commonly experienced in manual booking procedures.

The system also includes a **vehicle management module**, which stores complete information about every vehicle available for rental. Each vehicle record contains details such as the vehicle model, plate number, rental rate, seating capacity, current status, and maintenance schedule. Administrators can easily add new vehicles, update existing information, or temporarily deactivate vehicles undergoing maintenance. By maintaining updated vehicle records, the business can efficiently monitor fleet availability and reduce operational delays.

Another important component of the system is the **cash register and billing module**. Instead of manually calculating rental fees, insurance costs, penalties, and other charges, the system automatically computes the total amount based on the customer's reservation details. This minimizes mathematical errors and ensures that customers receive accurate billing information. After payment has been processed, the system generates a digital receipt that can be printed or stored electronically for future reference. Every transaction is automatically recorded in the database, making financial tracking faster and more organized.

The developed system also provides **payment management** capabilities. Administrators can record different payment methods, monitor payment status, and review previous transactions whenever necessary. Since all payment records are stored digitally, retrieving transaction history becomes significantly easier compared to searching through paper documents. This feature improves transparency and accountability in financial operations while supporting accurate bookkeeping.

4.3 System Development Methodology

The development of the **Calma Transportation Services: Cash Register and Reservation System** followed the **Agile Software Development Methodology**. Agile was selected because it provides a flexible and iterative approach to software development, allowing continuous improvement throughout the project. Instead of developing the entire system in a single phase, Agile divides the development process into smaller, manageable iterations. Each iteration focuses on developing specific features, gathering feedback, identifying issues, and implementing improvements before proceeding to the next stage. This methodology enabled the researchers to create a system that effectively addressed the operational needs of Calma Transportation Services while maintaining high software quality.

Throughout the development process, the researchers worked closely with the business owner to gather information about the existing reservation and payment procedures. Regular consultations were conducted to verify the system requirements and ensure that every feature accurately reflected the actual workflow of the business. Feedback collected during every development cycle was carefully evaluated and incorporated into succeeding revisions. This continuous collaboration helped minimize misunderstandings and ensured that the final system met both the technical requirements and business objectives.

The Agile methodology adopted in this study consisted of five major phases: **Requirements Analysis, System Design, System Development, System Testing, and Deployment**. Each phase contributed significantly to the successful completion of the system.

Requirements Analysis

The first phase involved gathering and analyzing the requirements needed for the development of the proposed system. The researchers conducted interviews and consultations with the owner and staff of Calma Transportation Services to understand the existing business processes, identify operational problems, and determine the features needed in the new system.

During this phase, the researchers examined how reservations were recorded, how payments were processed, how vehicle availability was monitored, and how customer information was managed. The researchers discovered several challenges associated with the manual system, including booking conflicts, inaccurate billing computations, difficulty retrieving customer records, delayed report preparation, and poor monitoring of vehicle maintenance schedules.

Based on the collected information, the researchers identified the functional and non-functional requirements of the system. Functional requirements included secure administrator login, customer registration, reservation management, vehicle management, automated billing, payment recording, receipt generation, maintenance scheduling, and report generation. Non-functional requirements included system security, usability, reliability, maintainability, and acceptable system performance. These requirements served as the foundation for the succeeding development phases.

System Design

After completing the requirements analysis, the researchers proceeded with the system design phase. This phase focused on planning the overall architecture, database structure, user interface, and workflow of the application before actual coding began.

The researchers designed the system using a client-server architecture in which users interact with the application through a web browser while the server processes requests and stores information in the database. Database tables were carefully structured to organize customer information, reservation records, vehicle details, payment transactions, maintenance schedules, and administrative accounts.

The user interface was designed with simplicity and ease of use in mind. Each page was arranged to allow administrators to perform daily tasks efficiently with minimal navigation. Input validation, organized menus, clear navigation, and responsive layouts were incorporated to improve the overall user experience.

Flowcharts, system flow diagrams, database relationships, and interface prototypes were also prepared during this stage to serve as guides during software development.

System Development

The development phase involved transforming the approved system design into a fully functional web-based application. The researchers developed the frontend using **React with TypeScript** to create an interactive and responsive user interface. The backend was developed using **Node.js**, which handled business logic, user authentication, reservation processing, payment computation, and communication with the database. **MySQL** served as the database management system for storing and managing all application data.

Each module of the system was developed incrementally following Agile practices. Major modules included user authentication, dashboard management, reservation management, vehicle management, automated billing, payment processing, maintenance monitoring, and report generation.

During development, the researchers continuously integrated new features while conducting regular debugging and code refinement. Whenever issues were identified, immediate corrections were made before additional modules were implemented. This iterative process helped improve software quality while reducing the possibility of major errors during later stages of development.

System Testing

Once the core features had been developed, the system underwent extensive testing to verify that every function operated correctly and satisfied the identified requirements. Different testing methods were performed to ensure system reliability, accuracy, security, and usability.

Functional testing was conducted to verify that each module performed its intended tasks correctly. Reservation creation, billing computation, payment recording, report generation, maintenance scheduling, and user authentication were individually tested before evaluating the system as a whole.

Integration testing ensured that all modules communicated properly with one another. For example, reservation records automatically updated vehicle availability, payment transactions generated receipts, and completed rentals were reflected in system reports.

User acceptance testing (UAT) was also conducted with the business owner and selected users. Participants evaluated the usability, functionality, accuracy, and overall performance of the developed system. Feedback obtained during this phase was documented and used to make final improvements before deployment.

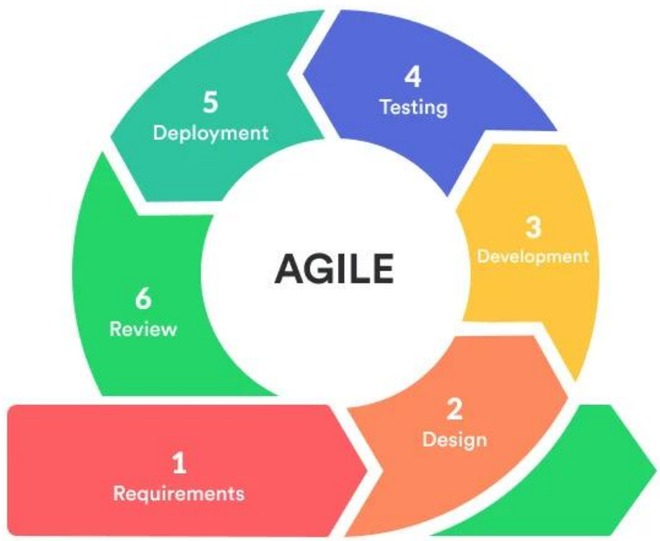
Deployment

The final phase involved deploying the completed system for actual use by Calma Transportation Services. Before deployment, the researchers performed final inspections to ensure that all modules functioned properly and that the database contained accurate information.

Administrative accounts were configured, sample operational data were entered, and users were oriented on the basic functions of the system. The researchers also demonstrated how to manage reservations, process customer payments, update vehicle records, generate reports, and monitor vehicles.

After deployment, the researchers continued to monitor system performance and addressed minor issues reported during the initial period of use. Additional adjustments were implemented whenever necessary to improve stability and usability. This continuous improvement process reflects the Agile philosophy of refining software based on user feedback even after implementation.

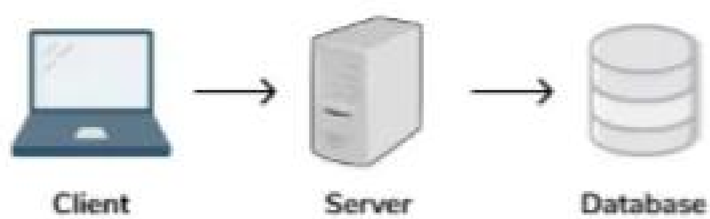
Overall, the Agile Software Development Methodology enabled the researchers to develop a reliable, efficient, and user-friendly Cash Register and Reservation System for Calma Transportation Services. The iterative nature of the methodology allowed continuous testing, evaluation, and enhancement throughout the project, resulting in a system that effectively addressed the operational challenges identified during the initial assessment.



4.4 System Architecture

The system architecture of the **Calma Transportation Services: Cash Register and Reservation System** follows a **client-server architecture**, which provides a centralized, secure, and efficient way of processing and managing data. This architecture enables the administrator to access the system through a web browser while the server handles application logic, processes requests, and communicates with the database. By separating the presentation layer, application layer, and data layer, the system becomes easier to maintain, update, and expand as the business grows.

The developed system consists of three primary components: the **client**, the **application server**, and the **database server**. These components work together to ensure that reservation records, customer information, payment transactions, vehicle data, and maintenance schedules are processed accurately and stored securely.



The **client side** serves as the user interface of the application. It was developed using **React with TypeScript** and allows authorized administrators to perform various operational tasks through a web browser. After logging into the system, the administrator can manage reservations, register customers, update vehicle information, process payments, monitor maintenance schedules, and generate operational reports. The interface was designed to be responsive, user-friendly, and easy to navigate, allowing users to complete tasks efficiently while minimizing errors.

The **application server** functions as the core processing unit of the system. Developed using **Node.js**, it receives requests from the client, validates user input, executes business rules, performs calculations, and manages communication between the user interface and the database. For example, when a reservation is submitted, the server verifies vehicle availability, calculates rental charges, records the transaction, and updates the vehicle's reservation status. The server also manages user authentication, ensuring that only authorized administrators can access the system and perform sensitive operations.

The **database server** uses **MySQL** to store and organize all system data. It maintains separate tables for administrator accounts, customer records, vehicle information, reservations, payment transactions, maintenance schedules, and generated reports. Through structured database relationships, the system can quickly retrieve and update information while maintaining data consistency and integrity. Proper database design also minimizes redundant information and improves overall system performance.

The interaction among these components begins when an administrator accesses the system through a web browser and submits a request, such as creating a reservation or processing a payment. The request is transmitted to the application server, where it is validated and processed according to the system's business rules. If database access is required, the application server communicates with the MySQL database to retrieve or update the necessary records. Once processing is complete, the server sends the results back to the client interface, allowing the administrator to immediately view the updated information. This communication process occurs in real time, enabling faster transaction processing and more accurate business operations.

To protect business data, the architecture incorporates several security measures. Administrator access requires user authentication through a secure login page before any system functions can be accessed. Input validation is performed on both the client and server sides to reduce invalid entries and help protect against malicious input. Sensitive information, such as user credentials, is securely stored in the database using password hashing techniques. Access to system functions is restricted to authorized users, ensuring that confidential business information remains protected.

The architecture also supports the integration of the system's major functional modules. The **Reservation Management Module** records customer bookings and automatically updates vehicle availability. The **Vehicle Management Module** maintains detailed information about each vehicle, including its rental status and maintenance history. The **Cash Register and Billing Module** automatically computes rental fees and generates digital receipts after successful payment. The **Maintenance Monitoring Module** tracks scheduled servicing and temporarily marks vehicles as unavailable whenever maintenance is required. Finally, the **Report Generation Module** compiles reservation, payment, maintenance, and vehicle information into organized reports that assist management in monitoring daily business performance.

One of the strengths of the client-server architecture is its scalability. As Calma Transportation Services expands its operations, additional features such as online customer reservations, mobile application support, SMS or email notifications, and digital payment gateway integration can be incorporated without requiring major modifications to the existing system architecture. This flexibility ensures that the developed system can continue supporting future business requirements while maintaining reliable performance.

Overall, the client-server architecture provides a stable and efficient foundation for the Cash Register and Reservation System. It enables secure communication between users and the database, automates essential business processes, reduces manual work, improves data accuracy, and supports faster decision-making through real-time information processing. By implementing this architecture, the developed system successfully meets the operational requirements of Calma Transportation Services while providing a scalable platform for future enhancements.

4.5 Hardware and Software Requirements

The successful implementation of the **Calma Transportation Services: Cash Register and Reservation System** requires appropriate hardware and software resources. These specifications ensure that the system performs efficiently during reservation processing, payment transactions, report generation, and database management while providing a reliable experience for both administrators and customers.

4.5.1 Hardware Requirements

Table 4.1 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 (10th Gen) or AMD Ryzen 3	Intel Core i5 (12th Gen) or AMD Ryzen 5 and above
Memory (RAM)	8 GB	16 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
Monitor	1366 × 768 Resolution	Full HD (1920 × 1080)
Internet Connection	20 Mbps	50 Mbps or higher
Printer	Thermal or Inkjet Printer	Thermal Receipt Printer
Mobile Device (Optional)	Android 10 / iOS 15	Android 13 or iOS 17 and above

The recommended hardware configuration provides sufficient processing power to handle reservation management, payment processing, receipt printing, vehicle monitoring, and report generation with minimal delays while ensuring smooth system performance.

4.5.2 Software Requirements

Table 4.2 Software Requirements

Software	Purpose
Windows 11	Operating System
Visual Studio Code	Source Code Editor
Node.js	Backend Development Environment
React with TypeScript (Vite)	Frontend Development
Bootstrap	User Interface Design
MySQL	Database Management System
XAMPP	Local Web Server and MySQL Management
Google Chrome	Web Browser for System Access and Testing
Git	Version Control
GitHub	Source Code Repository and Collaboration

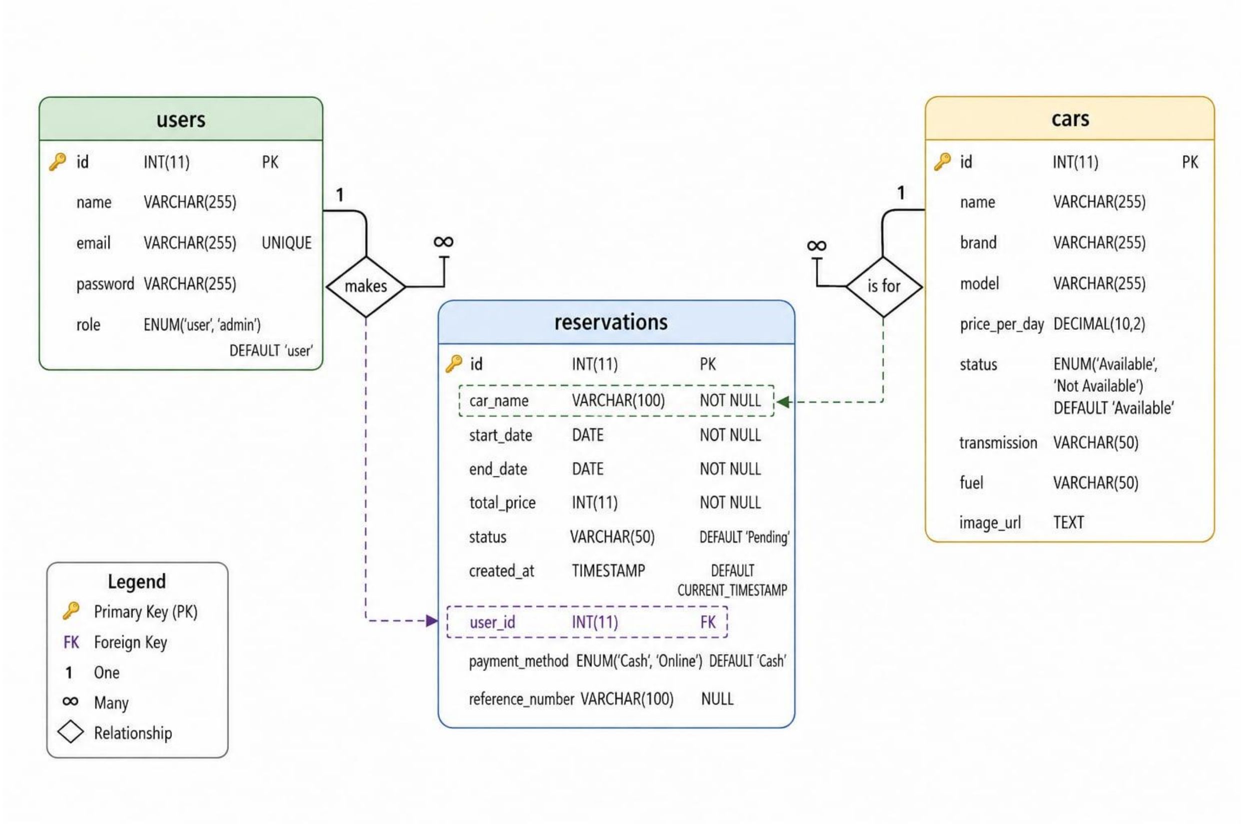
The selected software tools were chosen based on their reliability, compatibility, ease of development, and strong community support. Together, these technologies provide a stable environment for developing, testing, deploying, and maintaining the **Calma Transportation Services: Cash Register and Reservation System**, ensuring efficient reservation management, secure payment processing, accurate record keeping, and dependable system performance.

4.6 Database Design

The database serves as the foundation of the **Calma Transportation Services: Cash Register and Reservation System** by organizing and storing customer information, vehicle records, reservation details, payment transactions, cash register records, maintenance schedules, and administrative data. A well-designed database ensures efficient data storage and retrieval, minimizes data redundancy, and maintains the accuracy and integrity of information throughout the system's operations.

The researchers employed a **relational database model** using **MySQL** as the Database Management System (DBMS). Database normalization techniques were applied to eliminate unnecessary data duplication, maintain consistency among related tables, and establish proper relationships between entities such as customers, vehicles, reservations, payments, users, and maintenance records. Primary and foreign keys were implemented to enforce data integrity and ensure accurate linking of records across the database.

This database design enables the system to efficiently process reservations, manage vehicle availability, record payment transactions, generate sales and reservation reports, and support the daily operations of Calma Transportation Services while ensuring data security, reliability, and scalability for future enhancements.


users
id INT(11) PK
name VARCHAR(255)
email VARCHAR(255) UNIQUE
password VARCHAR(255)
role ENUM('user', 'admin') DEFAULT 'user'
1
makes
∞
reservations
id INT(11) PK
car_name VARCHAR(100) NOT NULL
start_date DATE NOT NULL
end_date DATE NOT NULL
total_price INT(11) NOT NULL
status VARCHAR(50) DEFAULT 'Pending'
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
user_id INT(11) FK
payment_method ENUM('Cash', 'Online') DEFAULT 'Cash'
reference_number VARCHAR(100) NULL
∞
is for
1
cars
id INT(11) PK
name VARCHAR(255)
brand VARCHAR(255)
model VARCHAR(255)
price_per_day DECIMAL(10,2)
status ENUM('Available', 'Not Available') DEFAULT 'Available'
transmission VARCHAR(50)
fuel VARCHAR(50)
image_url TEXT
Legend
Primary Key (PK)
FK Foreign Key
1 One
∞ Many
Relationship

CHAPTER 5: CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

This chapter presented the conclusions and recommendations derived from the development and implementation of the **Calma Transportation Services: Implementation of a Cash Register and Reservation System.** It summarized the significant findings of the study, determined whether the research objectives were achieved, and discussed the overall contribution of the developed system to the operations of Calma Transportation and Services. Furthermore, this chapter provided recommendations for future system enhancements, implementation strategies, and possible research directions to further improve the efficiency and sustainability of the developed system.

5.2 Summary of Findings

The study successfully developed and implemented a **Cash Register and Reservation System** for Calma Transportation Services to address the challenges associated with manual reservation processing, payment recording, customer information management, and report generation. The implementation of the system automated the company's daily operations, resulting in improved efficiency, accuracy, and reliability.

Based on the development, implementation, and evaluation of the system, the following findings were obtained:

- The developed system successfully computerized the reservation process, enabling administrators to record, update and monitor customer reservation efficiently.
- The system effectively monitored vehicle availability, thereby minimizing scheduling conflicts, and preventing double reservations.
- The integrated cash register module accurately computed rental fees, recorded customer payments, and generated electronic transaction records, reducing errors associated with manual computations.
- The centralized database securely stored customer information, vehicle records, reservation details, and payment transactions, allowing faster retrieval and better organizations of business records.
- The reporting module generated reservation summaries, payment reports, and transaction histories that supported management in monitoring business performance and making informed decisions.
- The developed system provided a user-friendly interface that simplified system navigation, reduced processing time, and improved the overall efficiency of daily operations.

Overall, the findings demonstrated that the developed system successfully achieved its intended purpose by replacing manual processes with an integrated computerized solution that enhanced operational efficiency and service quality.

5.3 Conclusion

Based on the findings of the study, it was concluded that the Calma Transportation services: Implementation of a Cash Register and Reservation System was a practical and effective solution to the operational challenges experienced by the company. The developed system successfully automated the reservation process, customer information management, payment processing, vehicle monitoring, and report generation, thereby improving the overall efficiency of business operations.

The implementation of the system significantly reduced the limitations of the previous manual process, including inaccurate record keeping, computation errors, misplaced documents, and reservation conflicts. The use of a centralized database ensured better organization, consistency, and security of business information while allowing authorized users to retrieve records quickly and accurately.

Furthermore, the developed system improved the quality of customer service by providing faster reservation processing, accurate payment recording, and efficient monitoring of vehicle availability. The integrated cash register module also strengthened financial accountability by maintaining complete and organized transaction records.

The objectives of the study were successfully achieved. The developed system demonstrated that integrating reservation management and cash register functionalities into a single web-based application effectively improved operational efficiency, reduced manual workload, enhanced customer service, and supported better business decision-making. The study also demonstrated the importance of adopting computerized information systems in modernizing transportation service operations.

5.4 Recommendations

Although the developed system successfully met its intended objectives, several recommendations were proposed to further improve its functionality, usability, and long-term sustainability.

System Improvements

Future versions of the system were recommended to include additional features that would further enhance business operations. These improvements included the integration of online payment gateways such as GCash, Maya, and credit or debit card payment options to provide customers with more convenient payment methods.

The researchers also recommended the implementation of SMS and email notification services to automatically send reservation confirmations, payment receipts, booking reminders, and vehicle return notifications.

A mobile application was likewise recommended to allow administrators and customers to access the system more conveniently through smartphones and tablets.

The integration of GPS vehicle tracking was also recommended to improve fleet monitoring and enhance vehicle management.

To strengthen data security, future versions of the system were recommended to implement multi-factor authentication, enhanced password encryption, comprehensive audit logs, and regular security updates.

Additional report export options, including PDF, Microsoft Excel, and CSV formats, were likewise recommended to improve documentation and report sharing.

Finally, the implementation of automated database backup and disaster recovery mechanisms was recommended to ensure data security and business continuity.

Future Research Directions

Future researchers were encouraged to expand the scope of the study by incorporating more advanced technologies into the system.

The integration of artificial intelligence and predictive analytics was recommended to forecast customer demand and vehicle utilization more accurately.

Researchers were also encouraged to develop business intelligence dashboards capable of providing deeper insights into customer behavior, reservation trends, and financial performance.

Future studies could also investigate the integration of cloud computing technologies to improve system scalability, accessibility, and remote data management.

Moreover, predictive vehicle maintenance modules could be incorporated to monitor maintenance schedules and minimize unexpected vehicle breakdowns.

Comparative studies involving similar transportation companies were likewise recommended to further evaluate the effectiveness and adaptability of computerized reservation and cash register systems in different operational environments.

Implementation Strategies

To maximize the effectiveness of the developed system, several implementation strategies were recommended.

Comprehensive user training should be conducted before full system deployment to ensure that administrators and employees became familiar with the system's features and functionalities.

Standard operating procedures should also be established to guide users in reservation processing, payment recording, report generation, and database management.

Regular software maintenance, database backups, and system updates should be performed to maintain system reliability, security, and optimal performance.

A qualified system administrator should likewise be assigned to monitor system performance, manage user accounts, perform routine maintenance, and address technical concerns.

Finally, continuous collection of user feedback was recommended to identify opportunities for further system enhancement and ensure that future improvements aligned with the operational needs of Calma Transportation Services.

5.5 Impact of the Study

The implementation of the **Calma Transportation Services: Implementation of a Cash Register and Reservation System** produced significant benefits for the company, its employees, customers, and future researchers.

For Calma Transportation Services, the developed system improved operational efficiency by automating reservation management, payment processing, and business record management. The system reduced manual workloads, minimized processing errors, and enabled faster retrieval of business information.

Employees benefited from simplified workflows, reduced paperwork, and improved productivity, allowing them to perform daily tasks more efficiently.

Customers also benefited from faster reservation processing, accurate booking information, organized payment transactions, and improved service quality, resulting in a more convenient and reliable customer experience.

From an academic perspective, the study demonstrated the practical application of information technology in solving real-world business problems within the transportation industry. The project served as a valuable reference for future researchers interested in reservation systems, cash register systems, transportation management, and business process automation.

Overall, the study highlighted the importance of digital transformation in improving operational efficiency, strengthening business management, and enhancing customer satisfaction through the implementation of modern information systems.

5.6 Summary

This chapter presented the conclusions and recommendations of the study. The findings demonstrated that the developed **Cash Register and Reservation System** successfully achieved its intended objectives by automating reservation management, payment processing, customer record management, vehicle monitoring, and report generation for Calma Transportation Services. The implementation of the system improved operational efficiency, minimized manual errors, strengthened data security, and supported more effective business decision-making.

Although the developed system successfully addressed the current operational requirements of the company, continuous enhancement was recommended to ensure long-term sustainability and adaptability to future technological developments. The proposed recommendations, including online payment integration, mobile application development, notification services, advanced reporting features, and intelligent analytics, were expected to further improve the system's overall functionality and effectiveness.

In conclusion, the study demonstrated that the implementation of a web-based **Cash Register and Reservation System** was an effective technological solution that modernized business operations, enhanced customer service, and supported the long-term growth and competitiveness of Calma Transportation Services.